

Spatial transcriptomics in the adult *Drosophila* brain and body

Reviewed Preprint

v2 • March 5, 2025

Revised by authors

Reviewed Preprint

v1 • January 16, 2024

Jasper Janssens, Pierre Mangeol, Nikolai Hecker, Gabriele Partel, Katina Spanier, Joy Ismail, Gert Hulselmans, Stein Aerts , Frank Schnorrer 

VIB-KU Leuven Center for Brain and Disease Research, KU Leuven, Leuven, Belgium • Laboratory of Computational Biology, Department of Human Genetics, KU Leuven, Leuven, Belgium • Aix Marseille University, CNRS, IBDM, Turing Centre for Living Systems, Marseille, France • VIB Center for AI & Computational Biology, KU Leuven, Leuven, Belgium

 https://en.wikipedia.org/wiki/Open_access

 Copyright information

eLife Assessment

This **important** study presents a method to visualize the location of the cell types discovered through single-cell RNA sequencing. The data allowed the authors to build spatial tissue atlases of the fly head and body, and to identify the location of previously unknown cell types. The data are **convincing** and appropriate, and the authors validate the methodology in line with the current state-of-the-art.

<https://doi.org/10.7554/eLife.92618.2.sa2>

Abstract

Recently, we have achieved a significant milestone with the creation of the Fly Cell Atlas. This single-nuclei atlas encompasses the entire fly, covering the entire head and body, in addition to all major organs. This atlas catalogs many hundreds of cell types, of which we annotated 250. Thus, a large number of clusters remain to be fully characterized, in particular in the brain. Furthermore, by applying single-nuclei sequencing, all information about the spatial location of the cells in the body and of about possible subcellular localization of the mRNAs within these cells is lost. Spatial transcriptomics promises to tackle these issues. In a proof-of-concept study, we have here applied spatial transcriptomics using a selected gene panel to pinpoint the locations of 150 mRNA species in the adult fly. This enabled us to map unknown clusters identified in the Fly Cell Atlas to their spatial locations in the fly brain. Additionally, spatial transcriptomics discovered interesting principles of mRNA localization and transcriptional diversity within the large and crowded muscle cells that may spark future mechanistic investigations. Furthermore, we present a set of computational tools that will allow for easier integration of spatial transcriptomics and single-cell datasets.

Introduction

Single-cell technologies have revolutionized biological research, allowing researchers for the first time to study complex tissues with unprecedented resolution (Klein et al., 2015 [↗](#)), leading to the creation of tissue-specific atlases (Schaum et al., 2018 [↗](#); The Tabula Sapiens consortium, 2022 [↗](#)). However, upon tissue dissociation into single cells, the spatial context of the cell in the organism and of the mRNAs within the cell is lost, since all data of the cell are compressed into one multidimensional data point. This is a significant drawback, as the localization of a cell type within a tissue can inform about the function of this cell (Tomancak et al., 2002 [↗](#)). Furthermore, many mRNAs show subcellular localization patterns that may have functional roles (Jambor et al., 2015 [↗](#); Lécuyer et al., 2007 [↗](#)).

This loss of information can be circumvented using spatially resolved transcriptomics (SRT). In SRT, spatial information of the RNA species is determined either through the use of spatially localized barcoded DNA arrays with or without the use of microfluidics, followed by sequencing (Rodrigues et al., 2019 [↗](#); Wang et al., 2022 [↗](#)), or by imaging using multiplexed rounds of single molecule fluorescence in situ hybridization (smFISH) methods. While barcoded sequencing approaches allow unbiased mapping of all transcripts, their spatial resolution is, in general, lower (Moses & Pachter, 2022 [↗](#)). Multiplexed smFISH identifies the position of each RNA molecule independently at a high spatial resolution, but it is limited in throughput to a few genes. Recently, technical breakthroughs have been presented to increase the number of genes to several hundred through the use of barcoding (Chen et al., 2015 [↗](#); Eng et al., 2019 [↗](#); Lubeck et al., 2014 [↗](#); Zhang et al., 2021 [↗](#)).

The adult fruit fly *Drosophila melanogaster* is a widely used model for cell biology, neurobiology, physiology and behavior (Sokolowski, 2001 [↗](#); Grenier & Leulier, 2020 [↗](#); Benton, 2022 [↗](#)). The recent completion of the entire adult fly cell atlas (FCA) (Li et al., 2022 [↗](#)), and the aging fly cell atlas (Lu et al., 2023 [↗](#)) provided an in-depth annotation of various cell types, however without information about the localization of these cells or their mRNAs. In addition, many single-cell clusters were left unannotated although marker genes were identified.

One tissue that would benefit greatly from SRT are the large polynucleated muscle cells. In tubular head and leg muscles, the nuclei are located in a central row surrounded by contractile myofibrils, while in fibrillar indirect flight muscles (IFMs), the nuclei are organized in multiple rows between the myofibril bundles (Luis & Schnorrer, 2021 [↗](#); Schönbauer et al., 2011 [↗](#)), distributed along the length of the large 1 mm long fibers (Spletter et al., 2018 [↗](#)). This organization is thought to optimize the mRNA distribution by minimizing transport distances (Bruusgaard et al., 2003 [↗](#)) and misalignment of the nuclei has been linked to severe myopathies in humans (Folker & Baylies, 2013 [↗](#)). While single nuclear (nc)RNA-seq allows studying of potentially different types of nuclei in the muscle, earlier studies have proposed classical FISH to study and describe mRNA localization in muscle fibers (Denes et al., 2021 [↗](#); Pinheiro et al., 2021 [↗](#)).

Here we applied Molecular Cartography (MC, Resolve Biosciences), a multiplexed single molecule hybridization method with submicron resolution (~136 nm), to create a spatial map of gene expression of the entire head and body of adult *Drosophila melanogaster* flies. While embryonic development had already been studied using SRT (Wang et al., 2022 [↗](#)), we present the first high-throughput smFISH-based SRT dataset of the various tissues of the adult fly. We compare the data with existing RNA atlases, using our spatial data to validate annotations in the body and identify the location of uncharacterized cell clusters in the brain. Furthermore, we reveal subcellular mRNA localization patterns in the flight muscle cells and regionalized RNA expression in the head,

some of which we confirm with classical FISH methods. Expression patterns of 100 genes in the brain and 50 genes in the body can be visualized interactively using TissUMaps (Solorzano et al., 2020) at www.spatialfly.aertslab.org.

Results

Creation of a comprehensive spatial dataset of the fly body and its head

In analogy to the FCA strategy (Li et al., 2022), we intended to map the spatial expression of a selected gene panel across the entire adult fly body and, in a separate set of experiments, across of the adult head. Based on expression data from the FCA (Li et al., 2022, Pech et al., 2024) we selected 50 genes for the fly body samples, and 100 genes for the head samples given the large heterogeneity of neurons. These genes were carefully chosen to label the most important known cell types of the adult fly, and to include some unknown ones, suggested from the atlas data. They cover a wide range of expression levels, from high to very low expressed genes (Table S1, Table S2, Figure 1 – figure supplement 1). Adult fly samples were frozen, sectioned, attached to slides and fixed (see Methods), after which they were profiled using MC (Figure 1a). Using this workflow, we detected on average 190,622 mRNA molecules for the head samples (min = 101,548, max = 260,989), ranging from 56 (*Poxn*) to 23,439 (*trio*) mRNA molecules per gene (Figure 1 – figure supplement 2A, B) and an average of 1.5 million mRNA molecules for the body samples (min=1,448,593, max=1,727,479), ranging from 112 (*MsR1*) to 510,619 (*Act88F*) mRNA molecules per gene (Figure 1 – figure supplement 3A, B).

For the head samples, we aimed to image as many different neuronal populations as possible. Thus, we performed coronal sections at different depths in the head (13 sections, 12 animals, mixed sex, Table S3), along the anterior-posterior axis (Figure 1B, Figure 1 – figure supplement 2A, C). For the body samples, sagittal sections were performed through the middle of the animal (5 sections from one male) (Figure 1C, Figure 1 – figure supplement 3A). These sagittal sections show only small structural differences in the abdomen; overall they are very similar to each other, showing the reproducibility of our data (Figure 1 – figure supplement 3C, see Methods).

Spatial transcriptomics allows the identification of cell types in the body

The genes for the body datasets were selected to cover a wide range of different cell types. Neurons in the central and peripheral nervous system were identified using expression of *elav* and *Syt1* (Figure 2A). The glial marker *repo* shows the location of glial cells across the body (Figure 2A). The expression of different trypsin isoforms is unique to the digestive system. Interestingly, α - and β -Trypsin show distinct patterns, with β -Trypsin localized to the inner side of the gut, while α -Trypsin is more distal, suggesting a subcellular localization of these trypsin isoforms coding mRNAs to apical or basal regions of the gut enterocytes (Figure 2A). Using the fat body marker *AkhR* and the oenocyte marker *FASN2*, different populations of fat tissue and oenocytes were identified in the abdomen of the fly at the expected locations. The hemocyte marker *Hml* shows distinct local enrichments in head, thorax and abdomen (Figure 2A). In addition, *LanB1*, which codes for LamininB1, an important component of the extracellular matrix present around many tissues (Yarnitzky & Volk, 1995), is widely produced in different cell types and not only in hemocytes. While co-expression of *LanB1* with *Hml* in hemocytes is detected as reported (Matsubayashi et al., 2017) (15.5%) (Figure 2a), surprisingly most of *LanB1* overlaps with epithelial cells (*grh* (13.4%) and *pain* (28.0%)) or muscle cells (*Mlp84B* (40.9%) and *Mp20* (25.4%)) (see Methods). Next, we used *esg* to mark adult stem cell populations. The expression of *esg* in our sections is mostly limited to the gut, matching its reported expression in the intestinal stem cells (Jiang & Edgar, 2011) and the somatic cyst stem cells (Sênos Demarco et al., 2022).

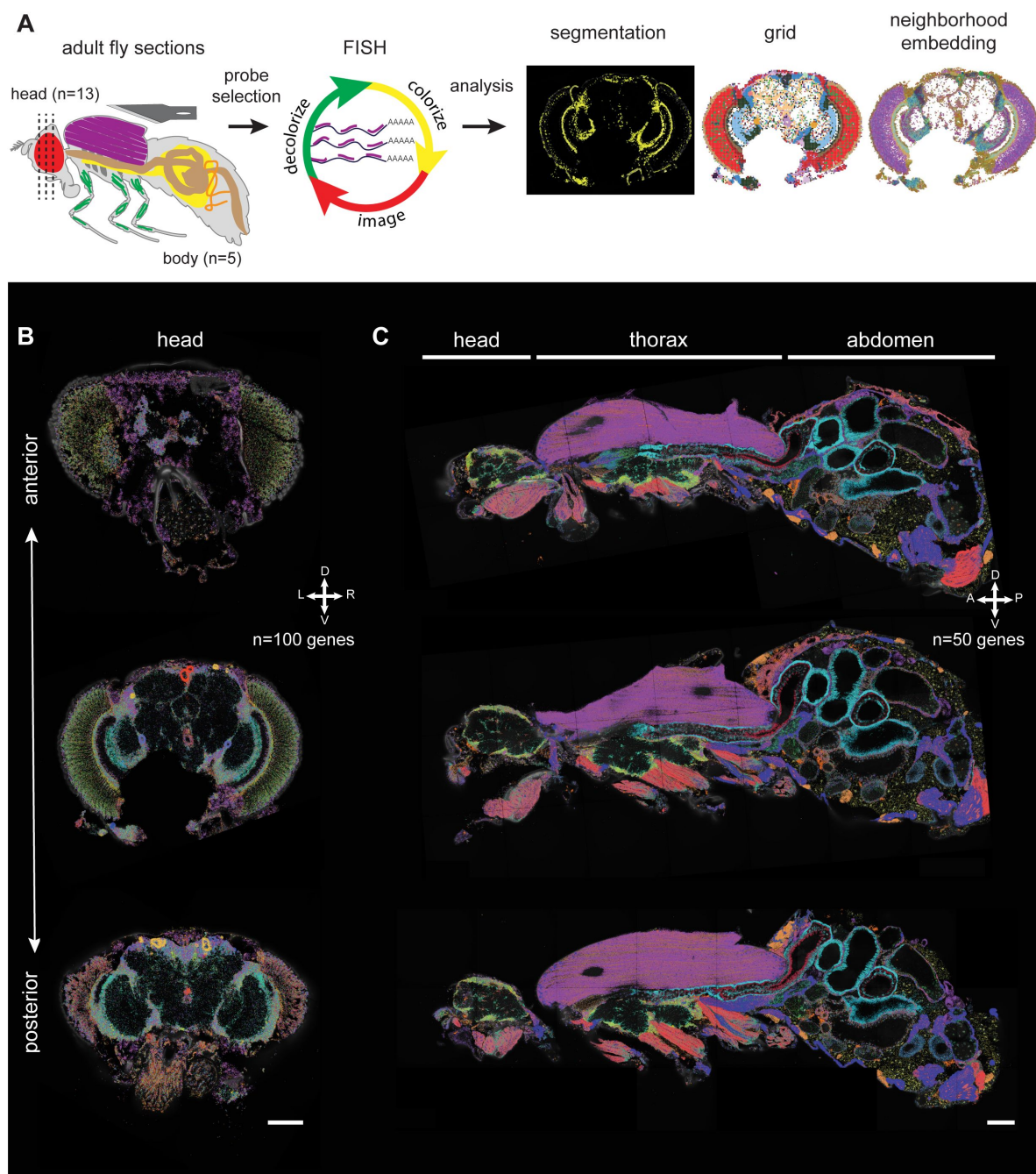


Figure 1.

Principal workflow of spatial fly transcriptomics.

(A) Overview of the spatial transcriptomics workflow: adult flies were sectioned, sections were analyzed with Molecular Cartography and data were annotated using cell segmentation, rasterization (i.e. grid) and neighborhood embedding (see Methods). (B, C) Three examples of adult head sections showing various positions in the brain along the anterior-posterior axis (B). Three examples of male whole-body sections taken from the same male (C). mRNAs from each gene are represented in a different color. The combination of colors reveals the different cell types. Scale bars represent 100 μ m. Background stain labels DAPI.

(15.5%) (**Figure 2a** [↗](#)), surprisingly most of *LanB1* overlaps with epithelial cells (*grh* (13.4%) and *pain* (28.0%)) or muscle cells (*Mlp84B* (40.9%) and *Mp20* (25.4%)) (see Methods)). Next, we used *esg* to mark adult stem cell populations. The expression of *esg* in our sections is mostly limited to the gut, matching its reported expression in the intestinal stem cells (Jiang & Edgar, 2011 [↗](#)) and the somatic cyst stem cells (Sênos Demarco et al., 2022 [↗](#)) (**Figure 2a** [↗](#)). Furthermore, *Mhc* and *sls* were used to label all muscle cells, while *TpnC4* and *Act88F* specifically labels the indirect flight muscles (IFMs) (**Figure 2A** [↗](#)). Interestingly, in the IFMs we also detect expression of VGlut and other neuronal markers, consistent with glutamatergic neuromuscular junctions (Schuster, 2006 [↗](#)) (Figure 2 – figure supplement 1). Finally, we used *grh*, *hth*, *svp* and *pain* to identify epithelial cells and their subtypes (**Figure 2A** [↗](#)). Together, these data show that our spatial transcriptomics data can identify the spatial location of most known large classes of cell types of the adult fly, while also detecting unexpected subcellular mRNA localizations in some cell types.

Molecular Cartography is highly specific

Most marker genes used in this study were selected based on FCA data to be highly specific for one particular cell type or present at one body location. Hence, their expression can be used to estimate the specificity of the mRNA detection method and determine false-positive rates of mRNA spots. For this specificity analysis, we chose to work with marker genes linked to muscle cells. We used the expression of *Act88F* and *TpnC4*, both of which are almost exclusively expressed in the IFMs (and possibly in muscles around the ejaculatory bulb) (Fyrberg et al., 1983 [↗](#); Agianian et al., 2004 [↗](#); Sarov et al., 2016 [↗](#)). We segmented the flight muscles (**Figure 2A** [↗](#)) and calculated the percentage of marker genes detected in the segmented area compared to the entire imaged area. We find that 99.1% of *TpnC4* and 99.7% of *Act88F* mRNA spots are detected in the expected regions of all body sections (**Figure 2A** [↗](#)), with negligible off-target signal. This highlights the specificity of the transcriptomics method and the reliability of detected mRNA signals with MC.

Building a spatial tissue atlas of the fly body

Cell types can often only be reliably identified by a combination of several marker genes. Therefore, we investigated co-expression signatures of known cell type marker genes in the spatial transcriptomics data. This may also help identify more rare cell types or specific cell states. To implement this idea on our spatial data, we computationally assigned every mRNA molecule to a 5 µm by 5 µm square in a square-grid fashion and summed the signal, creating pseudo-bulk samples. All squares in this grid were then scored for mRNA signatures of different cell classes and assigned to the highest scoring cell class (**Figure 2B** [↗](#), see Methods). This signature-based method annotated all major cell types or tissues of the adult reliably, including muscles, neurons, glia, fat and epithelial cells, oenocytes and the male reproductive organ (**Figure 2B** [↗](#)). In addition, cell populations with more sparse gene expression like the heart (combination of *Hand* and *tin*) or the male reproductive tract (combination of *Awh*, *eyg* and *svp*) were newly identified. In conclusion, the major cell types of the adult fly could be localized using combined sets of marker genes identified by their spatial proximity.

As a next step, we aimed to assign cell type locations without prior knowledge of their markers. To do so, we used all 132,642 different squares from each of the 5 body samples and performed clustering based on the counts of the 50 genes analyzed (see Methods). This led to the identification of 142 clusters that we visualized as a UMAP plot, separating muscle from neuronal and epithelial cells (**Figure 2C** [↗](#)). The spatial visualization of this unsupervised analysis of cell types confirmed our previous supervised annotation of cell types based on gene signatures (**Figure 2D** [↗](#)). In addition, it showed an unexpected presence of multiple clusters in the large flight muscle cells, suggesting subcellular mRNA patterns.

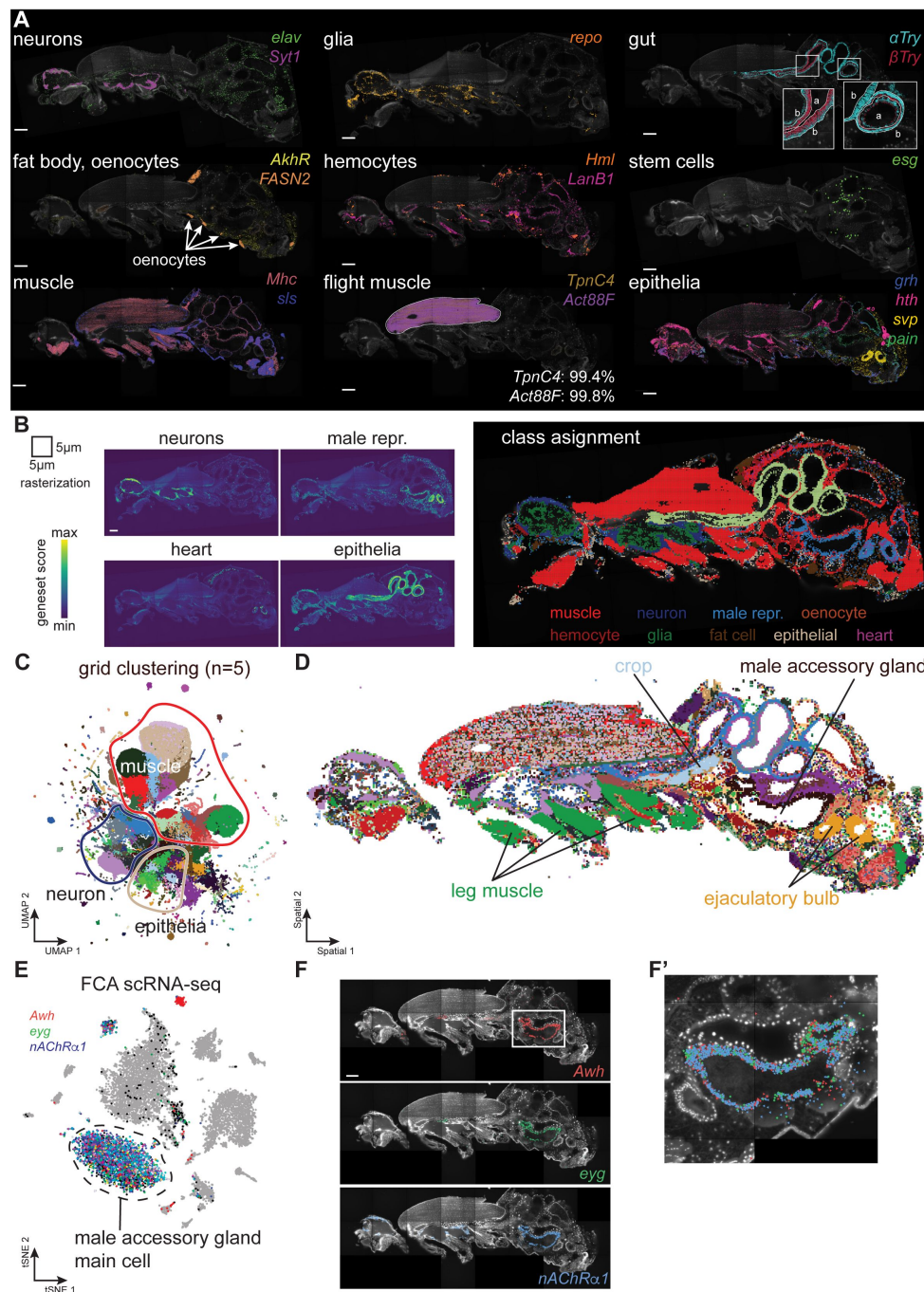


Figure 2.

Adult body cell types.

(A) Major cell types of adult males identified by marker genes. Scale bars represent 100 μm. Inset for gut shows zoom-ins of different regions (a: apical, b: basal). Inset for flight muscle, shows the percentage of marker gene molecules detected within the outlined area in the section shown. (B) Gene set scores for the main classes of cell types, quantified using 5 μm x 5 μm grid. Scale bar represents 100 μm. The class assignment shown on the right is based on maximum score across classes. Genes used: neurons (*elav*, *Syt1*, *Sh*, *acj6*, *ey*, *VAcHt*, *Gad1*, *VGlut*, *nAChRα7*); male reproductive system (*Awh*, *eyg*, *svp*); epithelia (*grh*, *aTry*, *βTry*, *hth*); heart (*tin*, *Hand*); muscle (*Mhc*); hemocyte (*Hml*); glia (*repo*); fat cells (*AkhR*); oenocytes (*FASN2*). (C) UMAP showing clustering of 5 μm x 5 μm grid spots. (D) Spatial location of grid clusters. (E) tSNE from male accessory glands from FCA showing expression of marker genes for main gland cells. (F) MC of the main gland cells marker genes highlights several defined populations of cells. Scale bar represents 100 μm. Detailed view shown in F'. Background stain labels DAPI.

Finally, we looked at the expression of genes in one particular organ to identify its different cell types. In the FCA, we had previously identified *Awh*, *eyg* and *nAChRa1* to label the male accessory gland main cells (**Figure 2E**). Our SRT data confirmed this co-expression, showing the labeling of the main gland cells in the abdomen (**Figure 2F, F'**).

Furthermore, we inspected whether co-expression of genes is maintained between SRT data and FCA snRNA-seq data, by calculating the correlation between genes in both datasets (see Methods). In general, gene-gene correlations between SRT and snRNA-seq match significantly ($r=0.69$, $p<1e-100$), although there are some small biases between the techniques (Figure 2 – figure supplement 2A, B). For example, *Act88F* is detected at higher numbers in SRT, compared to snRNA-seq, which may be explained with differential mRNA stability in the cytoplasm, since snRNA-seq is largely detecting nuclear located mRNAs and thus rather monitors the transcriptional activity of the cell. This indicates that in general SRT data corroborate results obtained from snRNA-seq. To study gene-gene relations in more detail, we devised an algorithm that calculates the colocalization of each gene with all other genes (Figure 2 – figure supplement 3A, B, see Methods). This colocalization matrix was then clustered to find groups of co-expressed genes. These co-expression signatures match with markers of the major cell types (Figure 2 – figure supplement 3C).

In conclusion, we present the first SRT dataset of the adult fly body, with high specificity and high reproducibility across body sections. We show how the SRT data can unambiguously identify the major cell types in both supervised and unsupervised techniques. Finally, SRT data can be used to describe gene-gene interactions.

Subcellular localization of mRNA in muscle cells

Unsupervised clustering revealed the presence of multiple spatial niches in the muscles, especially in the IFMs (**Figure 2C, D**). Muscle cells form polynucleated syncytia, with more than 500 nuclei in one IFM cell (Chaturvedi et al., 2017; Kaya-Çopur et al., 2021). Thus, spatial niches could represent heterogenous nuclei or subcellular mRNA localization patterns, resulting from directed mRNA transport or anchoring (Das et al., 2021). To investigate this in more detail, we segmented the three main types of adult muscles (head, leg and indirect flight muscles) to look at spatial gene expression in them.

Amongst the 50 genes we had selected were six genes that code for sarcomere protein components, four of which are specifically expressed in different muscle groups: *Act88F* and *TpnC4* in IFMs, *Act79B* and *TpnC41C* in leg muscles (**Figure 3A**). First, we investigated if mRNA species have different nuclear proximities by quantifying the distance of every mRNA molecule to its closest nucleus (see Methods, the localization precision of mRNAs is about 0.14 μm). This identified genes whose mRNAs are nucleus-enriched (*sls*) or nucleus-depleted (*TpnC4*) (**Figure 3B, C**, see also Figure 4 – figure supplement 1A). We found a similar difference in leg and head muscles with *sls* staying in proximity or within the nuclei, in contrast to the nuclei-depleted *TpnC41C* (Figure 3 – figure supplement 1A, B). Previously, mRNA distributions in cultured mammalian skeletal muscles were linked to mRNA sizes, with large mRNAs spreading further from the nuclei than small mRNAs (Pinheiro et al., 2021). This does not seem to apply to our selected genes, as *sls* mRNAs are significantly longer (isoform lengths: 15,263 to 56,489 nt) compared to *TpnC4* (longest isoform: 859 nt) or *TpnC41C* mRNAs (longest isoform: 1,880 nt).

Furthermore, the nuclear enrichment of *sls* mRNA is not homogeneous across all nuclei, particularly in the IFMs, where *sls* mRNA forms large assemblies around nuclei close to the muscle-tendon junctions (**Figure 4A-C**). We confirmed these results by localizing *sls* mRNA using hybridization chain reaction technology (HCR-FISH) (Choi et al., 2018) in thoraces from three adult males (**Figure 4D-F**, see Methods for details). We observed some small differences between MC and HCR, with *sls* mRNA enriched in broader regions around the muscle nuclei close to the tendons when detected with MC. This may be due to differences in fixation timing (pre-sectioning fixation for HCR, post-sectioning fixation for MC) or in image generation (direct imaging

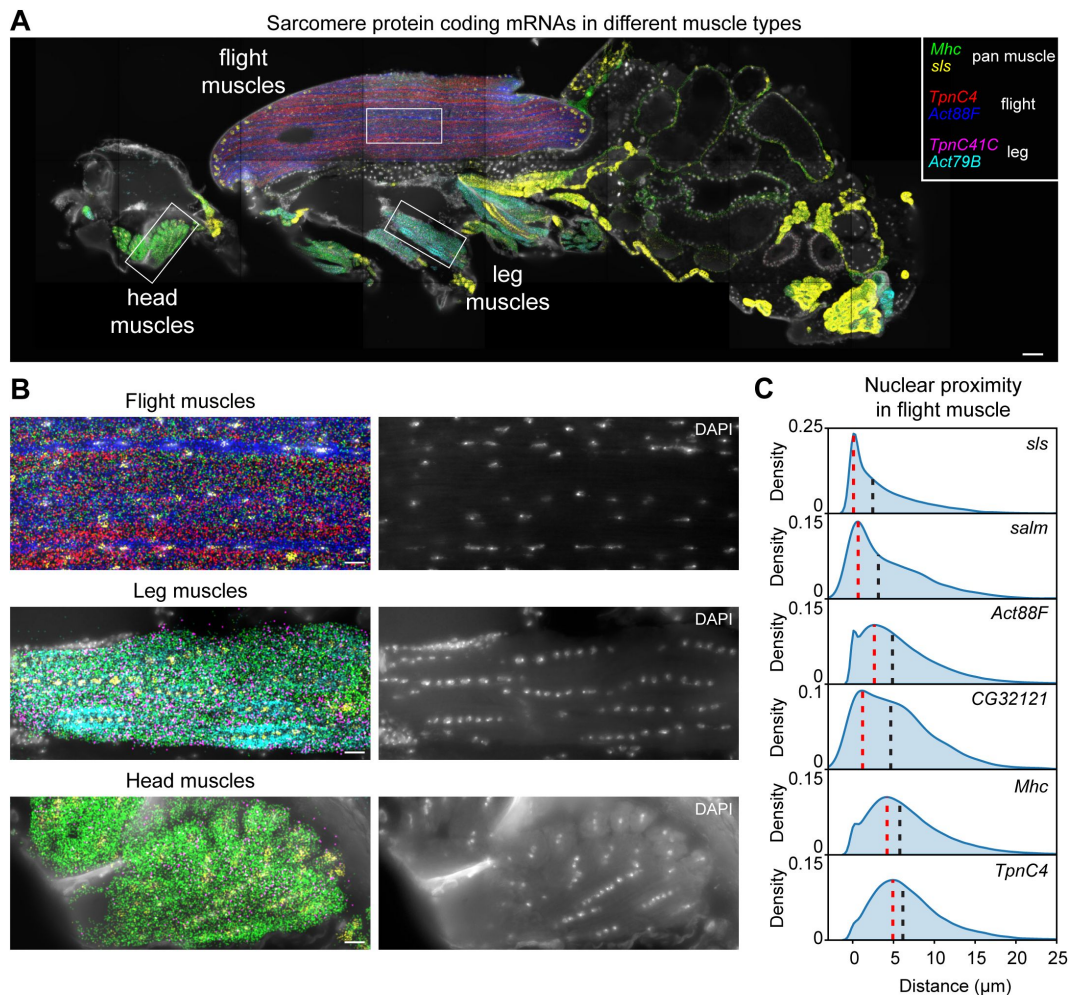


Figure 3.

Molecular Cartography shows mRNA-specific nuclear enrichment.

(A) Molecular Cartography (MC) visualization of marker genes of muscle subtypes. White boxes mark zoom-in regions shown in (B) and Figure 3 – figure supplement 1. (B) Zoom-in on flight, leg and head muscles. Left, labels are the same as in (A); right, DAPI-labelled nuclei. (C). Density plots show the distance of each mRNA molecule of the indicated genes to its nearest nucleus. Red dotted lines mark the peak density, and black dotted lines the median distance. Scale bars represent 50 μm in (A) and 10 μm in (B). Background stain labels DAPI.

of hybridization events in HCR vs detecting, localizing and combining multiplexed hybridization events in MC). Similarly enriched localizations of mRNA in muscles were reported in mammals close to the muscle-tendon junctions and the neuromuscular junctions (Dix & Eisenberg, 1990 [↗](#); Sanes et al., 1991 [↗](#)). In mammals, this local enrichment correlates with an accumulation of nuclei (Bruusgaard et al., 2003 [↗](#)), however, we do not detect a significant correlation between the *sls* enrichment at the IFM ends and the nuclear DAPI signal (Figure 4 – figure supplement 1) and thus speculate that the terminal nuclei express higher levels of *sls* RNA than the central muscle nuclei.

In addition to nucleus-enriched or -depleted mRNA species, MC revealed mRNAs coding for several sarcomere proteins to be concentrated in longitudinal stripes in the muscles. Most prominently in the IFMs, *Act88F* mRNA is enriched in the inter-myofibril space, where most nuclei, endoplasmic reticulum and Golgi are located, while *TpnC4* and *Mhc* mRNAs appear enriched in complementary stripes at the location of the myofibrils and mitochondria (Figure 5A, B [↗](#)). This suggests a spatial subdivision of the flight muscles that correlates with their intracellular architecture (Avellaneda et al., 2021 [↗](#); Luis & Schnorrer, 2021 [↗](#)). A similar division was found in leg muscles, with central stripes of *Act79B* mRNA close to the nuclei while *Mhc* and *TpnC41C* mRNAs were enriched in the myofibril regions (Figure 3 – figure supplement 1B). To validate these observations, we performed HCR-FISH. This partially confirmed the anticorrelated expression patterns between *Act88F* and *TpnC4* in IFMs: *Act88F* mRNA was found enriched in stripes along the rows of nuclei, consistent with the MC data. However, *TpnC4* mRNA was also found enriched in a few cases around nuclei (Figure 5C, D [↗](#), Figure 5 – figure supplement 1). Additionally, HCR-FISH detected that *Mhc* mRNAs are enriched in proximity to the nuclei and not in proximity to the myofibrils as suggested by MC (Figure 5E [↗](#)). Overall, these data suggest that the main observations obtained here with MC are reliable, however, a systematic validation of SRT techniques with classical FISH is recommended.

To conclude, we found three main types of mRNA localization in the IFMs: nuclei-enriched patterns, complementary striped bands and concentrations at the terminal nuclei, close to the muscle attachment sites. To our knowledge, none of these patterns had been identified before.

Spatial transcriptomics allows the localization of cell types in the head and brain

To investigate the localization of cell types in the adult *Drosophila* head, we investigated the 13 sections of different fly heads sectioned across the longitudinal direction, covering most regions of the fly head. The diversity between sections originating from different brain regions is represented by lower correlations between the samples (Figure 1 – figure supplement 2C). Using the marker genes *para* (neurons), *alrm* (astrocytes) and *ninaC* (photoreceptors), we were able to annotate different classes of cells (Figure 6A [↗](#)). *ninaE*, a second photoreceptor marker was too highly expressed, leading to optical crowding and was removed from the analysis.

First, we made use of our annotated scRNA-seq atlas of the fly brain (Davie et al., 2018 [↗](#); Li et al., 2022 [↗](#); Pech et al., 2024 [↗](#)), using colocalization of marker genes to identify cell types with known anatomical locations. We used the expression of *C15*, *acj6*, *Oaz*, *caup* and *unpg*, markers for olfactory projection neurons (OPNs). OPNs relay information from the antennal lobe to downstream processing centers, including the mushroom body and the lateral horn. The co-expression of OPN markers is only detected in the most anterior samples, as expected (Figure 6B [↗](#)). Next, we looked at *repo*, *moody*, *Indy*, *Vmat*, all of which are markers for perineurial glia that form the blood-brain barrier (BBB), and found them expressed at the periphery of the brain, consistent with the known location of the BBB (Figure 6C [↗](#)).

We compared gene-gene relationships, by calculating gene-gene correlations in MC and the snRNA-seq brain data (Figure 6 – figure supplement 1A, B). In general, we find a strong match ($r=0.68$), however, there are also several mismatches. While *pros-dati* co-expression is conserved between both modalities (Figure 6 – figure supplement 1C), co-expression of several other gene

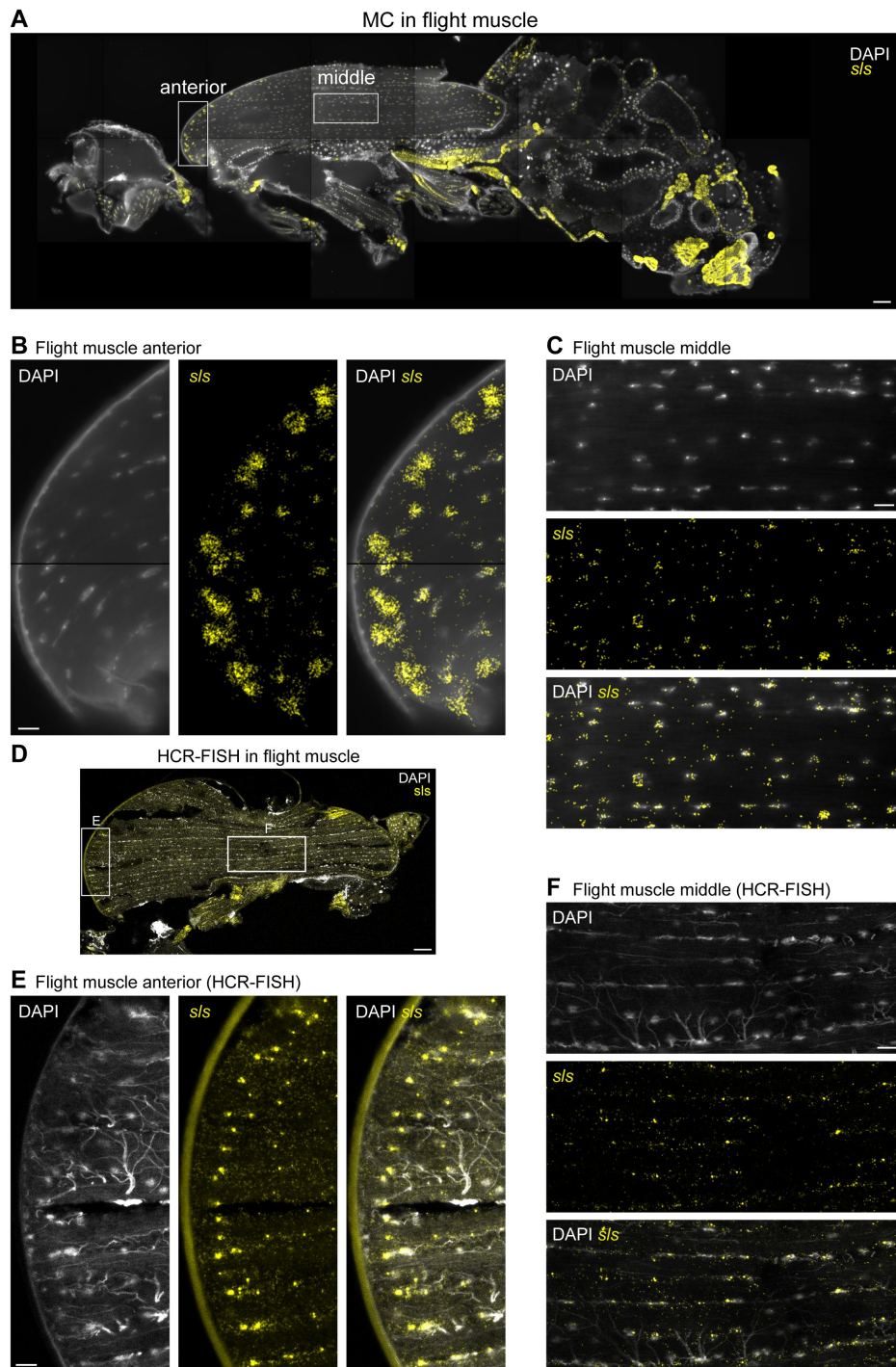


Figure 4.

***s/s* mRNA show nuclear enrichment with increased concentration close to muscle-tendon junctions.**

(A) Molecular Cartography visualization of *s/s* mRNA (yellow) as pan muscle maker (same section as in [Figure 3](#)). White boxes mark zoom-in regions shown in (B) and (C). (B, C) Zoom-in on indirect flight muscles showing colocalization of *s/s* mRNAs and DAPI stained nuclei in anterior (B) or central (C) regions of the flight muscle. (D) HCR-FISH imaging of *s/s* mRNA in an adult thorax. White boxes mark zoom-in regions shown in (E) and (F). (E, F) Zoom-ins on flight muscle at anterior (E) or middle (F) regions. Note the *s/s* mRNA and DAPI co-localization (tracheal cells show background stain in the DAPI channel). Scale bars represent 50 μ m in (A) and (D), and 10 μ m in (B), (C), (E), and (F). Background stain labels DAPI.

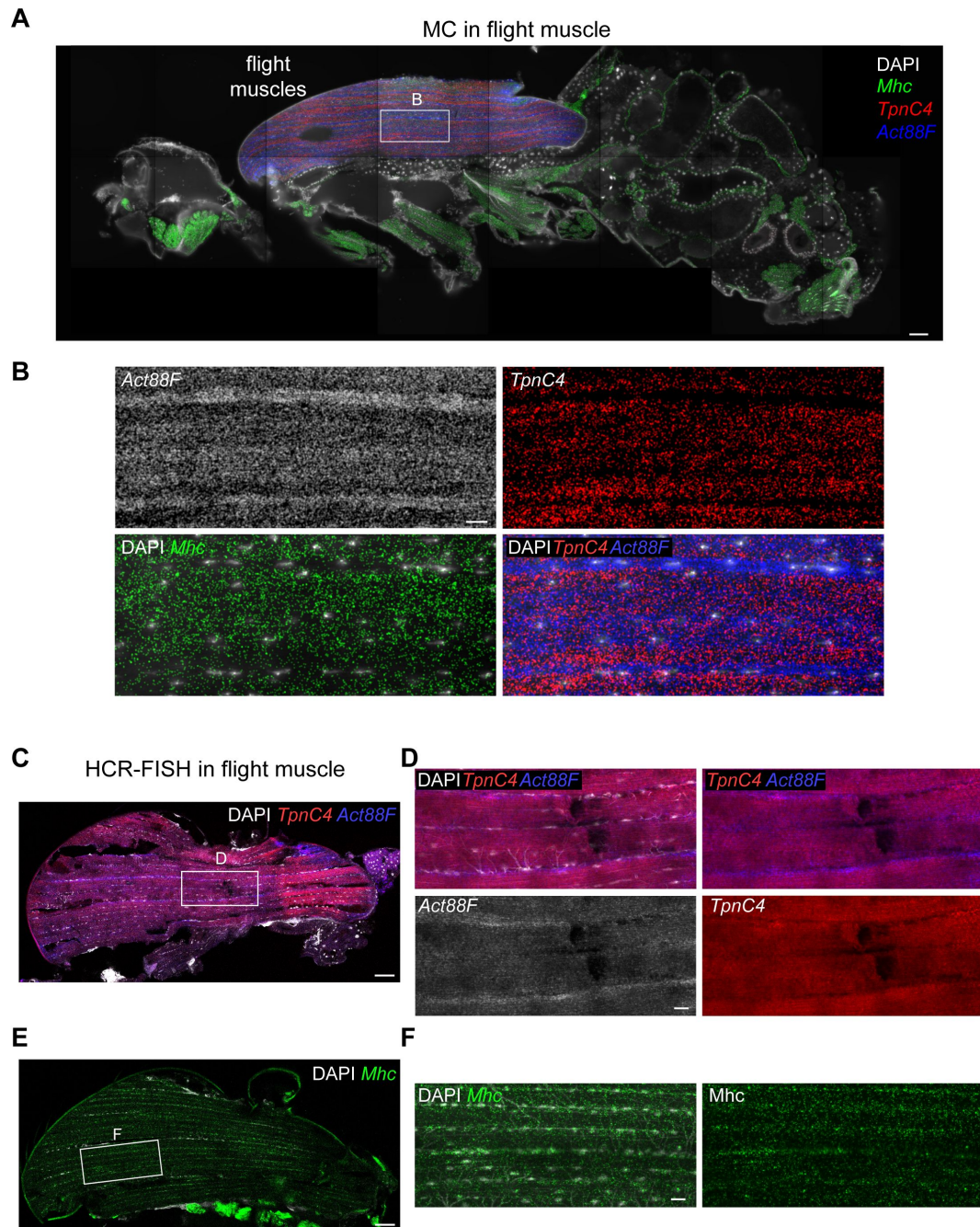


Figure 5

– *TpnC4*, *Act88F*, and *Mhc* mRNAs flight muscle patterning.

(A) Molecular Cartography visualization of *TpnC4*, *Act88F* and *Mhc* mRNAs on an adult fly section (same section as in **Figure 3**). The white box marks the zoom-in region shown in (B). Zoom-in on the flight muscle showing the striped patterns of *TpnC4*, *Act88F* (blue in overlay), and *Mhc* mRNAs in the indirect flight muscle. (C) HCR-FISH of *TpnC4* and *Act88F* mRNAs in an adult thorax. The white box marks zoom-in the region shown in (D). (D) Zoom-in on the flight muscle of HCR-FISH labelling *TpnC4* and *Act88F* mRNAs (same section as in **Figure 4D**). (E) HCR-FISH of *Mhc* mRNAs in an adult thorax. The white box marks the zoom-in region shown in (F). (F) Zoom-in on the flight muscle of HCR-FISH labeling *Mhc* mRNAs. Scale bars represent 50 μm in (A), (C), (E), and 10 μm in (B), (D), (F). Background stain labels DAPI.

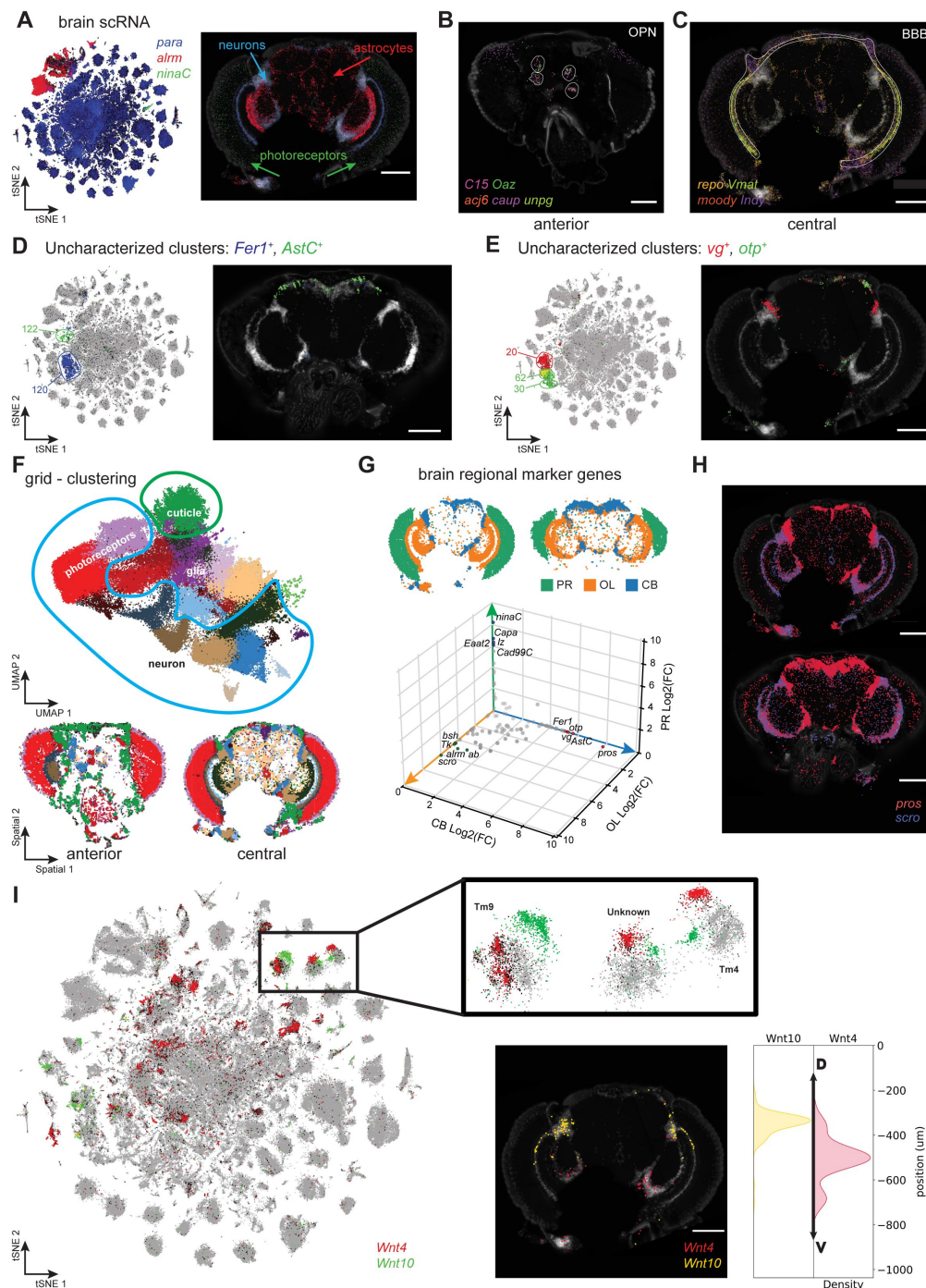


Figure 6.

Adult head cell types.

(A) tSNE showing expression of photoreceptor (*ninaC*), neuronal (*para*) and glial (*repo*) markers (left). Molecular Cartography of the same marker genes (right). (B, C) Molecular Cartography of marker genes for olfactory projection neurons (OPN), in an anterior head slice (B) and of perineurial glia of the blood brain barrier (BBB) in a more central brain slice (C). (D, E) Using Molecular Cartography to localize uncharacterized clusters found in scRNA-seq data. (F) UMAP showing clustering of 5 μ m x 5 μ m grid spots (top). Spatial location of grid clusters in the brain (bottom). (G) Differential expression of central brain (CB), optic lobe (OL) and photoreceptor regions (PR). (H) Molecular Cartography of *pros* and *scro* in the brain. (I) tSNE showing split in optic lobe clusters by expression of *Wnt4* and *Wnt10*. Insert shows Molecular Cartography of *Wnt4* and *Wnt10*, spatially localized in ventral and dorsal brain regions, respectively. Scale bars represent 100 μ m. Background stain labels DAPI.

pairs is only detected in one modality (Figure 6 – figure supplement 1D, E). For example, we only found widespread expression of *Vmat* in the BBB (stronger near the optic lobe), but little or no expression in the central brain, where most dopaminergic neurons should be located (Figure 6 – figure supplement 1E). Additionally, *Vmat* expression does neither overlap with DAPI-stained nuclei nor with *DAT* (marker for dopaminergic neurons), suggesting an mRNA transport mechanism to locate selected mRNAs away from the cell body (Figure 6 – figure supplement 1E). Alternatively, glial nuclei, which are much smaller than neuronal nuclei (Mu et al., 2021 [DOI](#)), might not all be detected with our methodology. Similarly, the neuropeptides *Ilp2* and *Pdf* show only a weak overlap with the DAPI staining, instead forming circular patterns (Figure 6 – figure supplement 1F). Thus, MC can recapitulate the location of known cell types in the fly brain by using established marker genes and additionally identify expression at novel locations for some of these.

Compared with other organs, the cell type diversity of the fly brain is extremely complex. A key challenge in scRNA-seq is the annotation of clusters to cell types. In our scRNA-seq dataset, there are 188 distinct clusters, of which only 83 are annotated today. Recent efforts to map the fly brain through EM and connectomics studies have identified 8,453 morphological cell types (Dorkenwald et al., 2024 [DOI](#); Schlegel et al., 2024 [DOI](#)). SRT can provide a bridge between scRNA-seq studies and morphology-based studies. As such, it becomes possible to annotate unknown clusters based on the spatial localization of their marker genes. To pave the way towards such integration, we focused on unannotated clusters in our brain dataset. It is important to note that current neuronal nomenclature is based on neuropils (axons and dendrites) and not on the location of the neuronal nuclei. Therefore, we used the established neuropil nomenclature to describe the location of the detected nuclei, but this does not necessarily mean that the axons of these cells also project there.

To identify the location of uncharacterized clusters from the FCA in the brain, we started with *Fer1* expression, as it is a strong marker for cluster 120 in our snRNA dataset (Li et al., 2022 [DOI](#)). Using our MC data, we found that these cells are located in the central brain, near the ventral gnathal ganglion and saddle, and the superior intermediate protocerebrum (Figure 6D [DOI](#)). Furthermore, we show that *AstC* expression (cluster 122) is limited to the dorsal part of the central brain, consistent with its expression in DN1 and DN3 neurons (Díaz et al., 2019 [DOI](#)) (Figure 6D [DOI](#)). Next, we investigated the expression of *otp* (clusters 30 and 62), which we found marks nuclei close to the lateral horn and the superior neuropils (Figure 6E [DOI](#)). Finally, we determined the expression of *vg* (cluster 20) to be limited to the boundary region between the optic lobe (OL) and the central brain (CB) (Figure 6E [DOI](#)). These results confirm that combining snRNA-seq with SRT technologies can indeed identify defined locations of formerly uncharacterized cell clusters in the fly brain.

Next, we wanted to find genes enriched to specific spatial locations in our brain MC data. To do so, we assigned mRNA molecules to spots using a similar 5 µm resolution grid as done above for the body data. By performing unsupervised clustering using all mRNA localizations from all datasets together, we identified 23 clusters (Figure 6F [DOI](#)). Different slices display a different cluster composition, consistent with our aim for sectioning along the brain's A-P axis (Figure 6 – figure supplement 1G). All neuronal clusters were selected using *para* expression and annotated to either optic lobe (OL), central brain (CB) or retina (photoreceptor, PR) based on their location. By performing differential expression between these domains, we identified spatially relevant genes (Figure 6G [DOI](#)). As expected, photoreceptor genes such as *ninaC* and *lz* were strongly enriched in the retina. While many neuronal genes show similar expression between the OL and CB, *pros* (CB) and *scro* (OL) expression creates a strong boundary clearly demarcating the two anatomical regions (Figure 6G, H [DOI](#)), corroborating earlier findings (Davie et al., 2018 [DOI](#); Yoo et al., 2020 [DOI](#)). This shows that SRT alone can be used to identify specific markers for the major brain cell types or regional domains, without any prior knowledge.

Previously, it was shown that some scRNA-seq clusters in the OL could be divided into ventral *Wnt4* and dorsal *Wnt10* positive subclusters (Han et al., 2020 [↗](#); Özel et al., 2021 [↗](#)). Using *Wnt4* and *Wnt10* mRNA localizations we confirmed these findings and showed that, although expression is sparse, *Wnt4* and *Wnt10* mRNAs are strongly enriched in the ventral and dorsal regions of the brain, respectively (Figure 6I [↗](#)).

Altogether, our MC data of the fly brain allows the identification of clusters and cell types. We also showed how SRT can be used to identify and localize unknown clusters of scRNA data, serving as a potential link between morphology data and single-cell data. Finally, SRT can be used for the discovery and confirmation of regional marker genes by performing region-based differential expression.

Automated cell type annotation of SRT using scRNA-seq data

One of the key prospects for SRT, is to be able to identify unlabeled clusters and infer their spatial location. This requires integration of SRT data with scRNA-seq data. Multiple methods have been developed for this purpose so far, including Tangram (Biancalani et al., 2021 [↗](#)) and SpaGE (Abdelaal et al., 2020 [↗](#)).

Here we used and compared three different approaches to represent SRT data: 5 μ m-spaced grid rasterization, neighborhood embedding, and nuclei segmentation (Figure 7A [↗](#)). The grid and neighborhood embedding are both spatially unaware of the cell's location. In the grid method, automatic segmentation takes place over the entire sample by grouping mRNA molecules together every 5 μ m, while neighborhood embedding is segmentation-free and models every mRNA molecule independently. This makes it possible to also visualize spatial patterns at subcellular resolution, but leads to a very large dataset and heavy computational load (Partel & Wählby, 2021 [↗](#)). While the grid-based approach can be run in several hours, neighborhood embedding takes several days. Finally, we segmented the nuclei and assigned mRNA molecules to each nucleus. While this is the most intuitive method, several challenges occur in the fly head. To start, different densities of nuclei require imaging with different parameters to visualize both sparse and dense nuclei regions. In addition, the dense nuclei regions at the edge of the OL and CB are very difficult to segment with normal imaging techniques, with a strong overlap of cells and different cell types (Figure 7B [↗](#)).

To compare label transfer methods we applied lasso regression, Tangram and SpaGE. We found that regression-based integration in the grid method and SpaGE in the neighborhood embedding method can match the blood-brain barrier and the chiasm glia in the optic lobe very well, together with astrocytes and ensheathing glia in the central brain (Figure 7C [↗](#)). In contrast, the nuclei segmentation method performs poorly for matching glial subtypes, since most of the glial nuclei are not detected, leading to the removal of most glial mRNAs (Figure 7C [↗](#)). Therefore, we also ran Tangram on the grid-quantification, leading to the detection of different glial types (Figure 7 – figure supplement 1A-D). This shows a disadvantage of nuclei-aware methods compared to methods that take all mRNA spots into account.

In the OL, all methods can accurately identify neurons from the different layers. In the CB, both the grid-based and the segmentation methods are able to locate the peptidergic neurons (insulin-producing cells, Pdf-neurons) in the correct location, while neighborhood embedding fails to retrieve these (Figure 7D, E [↗](#)). However, Tangram also labels some non-peptidergic segmented nuclei as insulin-producing unless thresholds for mapping scores are manually adjusted to reflect the different proportions of cell types that are present in the MC data (Figure 7 – figure supplement 1E). In a more posterior CB sample, nuclei segmentation and grid methods are able to identify the Kenyon cells of the mushroom body. These are also found in the neighborhood embedding, but the signal is noisier (Figure 7E [↗](#)). Neighborhood embedding likely performs poorly on cell types that are determined by high expression of only 1 gene (e.g. neuropeptides), since one gene only has a marginal effect on the neighborhood.

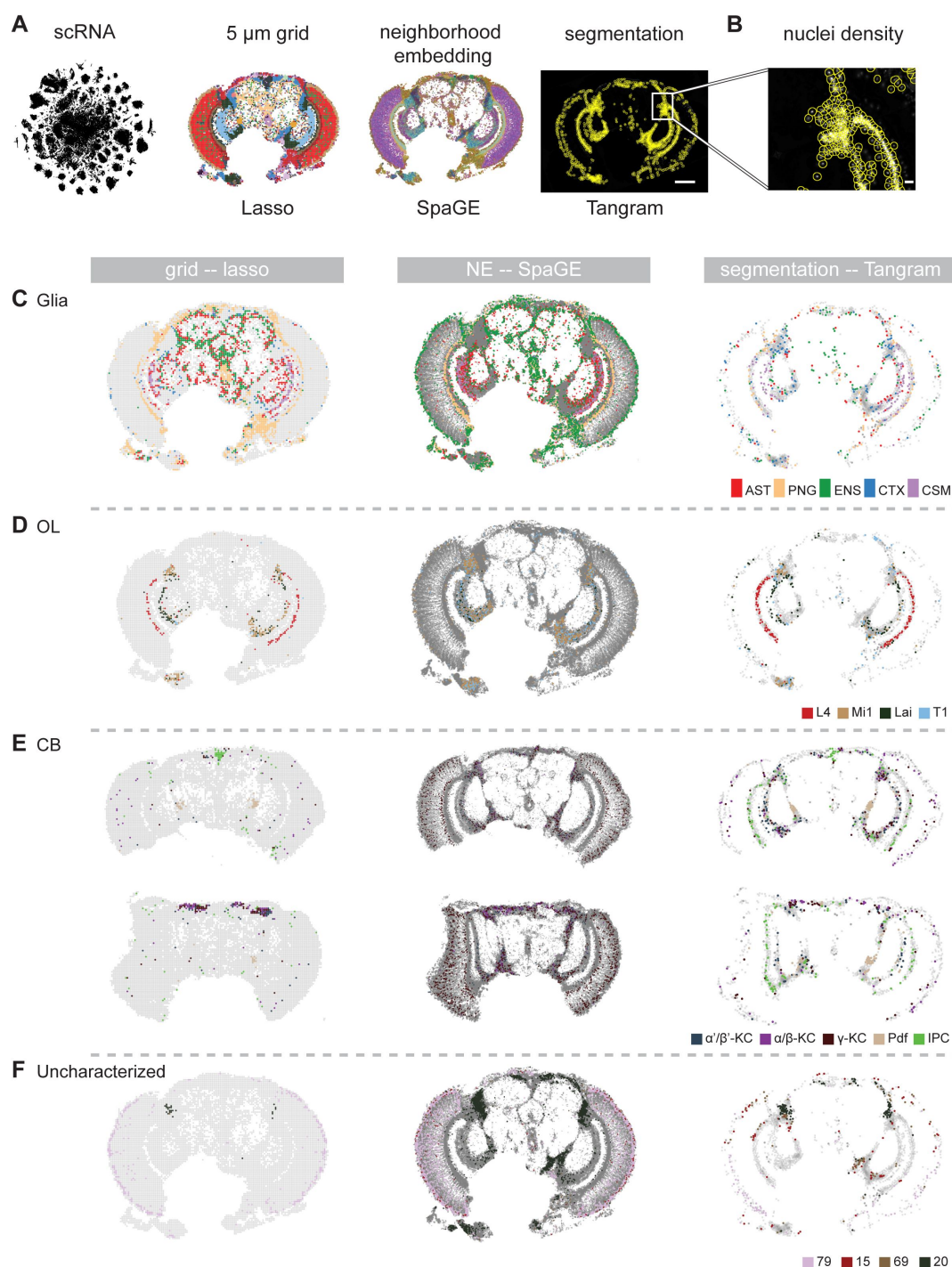


Figure 7.

Comparison of different techniques for annotating the adult head samples.

(A) Overview of different spatial analysis methods which were used to annotate Molecular Cartography with labels from single-cell RNA-seq: grid-based, neighborhood embedding and nuclei segmentation. Scale bar represents 100 μ m. (B) Zoom-in on a high-density region with corresponding segmented nuclei. Scale bar represents 10 μ m. (C-F) Comparison of annotation of spatial data with single-cell RNA-seq for three different quantification methods. Grid-based squares/neighborhoods/nuclei are colored based on matching single-cell clusters for (C) glia, (D) optic lobe, (E) central brain and (F) four uncharacterized clusters. In (E) two brain slices are shown at different depths: central (top) and posterior (bottom). NE: neighborhood embedding.

Finally, we investigated how the three approaches perform in linking uncharacterized scRNA-seq clusters to spatial locations. The grid and neighborhood embedding methods spatially localized two uncharacterized clusters, while the nuclei segmentation method spatially localized four clusters (**Figure 7F**). Cluster 20, which was marked by *vg* expression, is found by each technique (**Figure 6E**, **Figure 7F**), same as cluster 79. Cluster 79 locates to the outside of the retina, fitting with high expression of *Cpr72Ec*, a structural protein for the eye lens, which is expressed by interommatidial pigment cells (Stahl et al., 2017). In contrast to the other methods, Tangram detected a ventral-dorsal division in the retina, with cluster 15 matching to the dorsal part. This mapping is driven by the expression of *mirr*, a gene known to be expressed in the dorsal half of the eye (McNeill et al., 1997) (Figure 7 – figure supplement 1F). Tangram also maps cluster 69, mostly by expression of *caup*. Like its Iroquois family member *mirr*, *caup* is also detected in the dorsal half of the eye, however it shows additional expression in various locations in the central brain (Figure 7 – figure supplement 1G). Cluster 69 represents multiple subclusters in scRNA-seq data, and similarly it maps to multiple locations in the MC data. A larger gene panel will help to dissect and locate the subclusters in more detail in the future.

To conclude, we found that using a 5 μ m by 5 μ m grid leads to similar clustering results as performing a computationally heavy neighborhood embedding analysis. However, the neighborhood embedding does increase the resolution to single mRNA spots, leading to a better visualization and a higher spatial accuracy. The nuclei segmentation method is limited in the fly brain by the inability of the DAPI signal to resolve overlapping individual nuclei in very densely packed brain regions. While Tangram was the best method to identify unknown clusters, it also led to false-positive labeling of multiple cell types, leading to the necessity of accurate thresholding. We find that using a simple regression model works as well or even better as designated methods when marker genes are used, while running much faster. The designated methods might show improved performance when using larger gene panels. Therefore, we suggest relying on a consensus of different methods to optimally integrate scRNA-seq data with SRT.

Discussion

Here we present the first SRT dataset of the adult fly brain and body using a highly multiplexed single molecular in situ hybridization technique Molecular Cartography (MC). Most of the published classical mRNA in situ hybridizations (Tomančák et al., 2002; Lécuyer et al., 2007; Jambor et al., 2015) as well as a recent high throughput FISH study used the more easily accessible fly embryos (Wang et al., 2022). We expanded SRT here to the more complex anatomy of the adult fly, which requires sectioning and large data sampling, both at the microscope and at the computer. Using a 50-genes panel in the body, all major cell types in the adult male fly were identified. Furthermore, expression of key markers was sufficient to locate rare cell populations in the adult body. For example, expression of *Hml* showed the location of resident hemocytes, while *esg* was used to mark stem cells in the reproductive and intestinal tracts. We further showed a high concordance with published snRNA-seq data using several measures of co-expression (Li et al., 2022).

While SRT based on smFISH in mice (Yao et al., 2023) and humans (Fang et al., 2022) is currently limited to a single tissue, in flies both the brain and the whole-body can be sliced and imaged, allowing an unprecedented whole body view. This enabled us to investigate the specificity of marker genes in the context of all tissues. We found that many markers are indeed highly specific to certain cell types, as we quantified for *Act88F* and *TpnC4* for the flight muscles (>99% specificity), which is consistent with previous benchmarks of MC in human and mouse cells (Groiss et al., 2021). Still, some marker genes showed additional unexpected expression (e.g., expression of *tin* and *TpnC4* in the ejaculatory bulb muscles), which was not detected in snRNA-seq. Additionally, we observed widespread expression of *LanB1* in various cell types, beyond just

hemocytes. As such, SRT can provide a more sensitive readout, including the spatial localization, and may inspire which other tissues might be interesting to investigate in the future when studying a particular gene function.

In the head, we used a 100-genes panel and showed that this selection is sufficient to identify the major classes of cell types in the brain, including some very specific neuronal cell types like peptidergic neurons, Kenyon cells and different OL neurons. Furthermore, by using informative markers enriched in previously uncharacterized clusters from single-cell RNA-seq data, we were able to pinpoint the nuclei in the fly brain that express these markers. This allowed us to map a series of previously uncharacterized cell clusters to specific brain regions. The next challenge lies in connecting these nuclei to the neuronal connectome using connectomics studies (Schlegel et al., 2024 [DOI](#)), in order to unambiguously identify the individual cell types. Indeed in *Drosophila*, neuronal nomenclature and cell typing focus on projections, instead of nuclei localization. We believe that future large-scale SRT data can serve as a bridge between different modalities, similar to the Rosetta stone, to link scRNA-seq atlases with spatial information, including connectomics in the brain.

Performing SRT on the adult head led to two additional challenges. The first issue is technical, since the small area of the head made it difficult to achieve homogeneous adhesion to the slides, leading to significant deformation of the samples. Secondly, the small and dense *Drosophila* brain nuclei are hard to segment (average size 5 cubic micrometers (Mu et al., 2021 [DOI](#))). A possible solution would be to use SRT-compatible expansion microscopy (Fan et al., 2023 [DOI](#); Pownall et al., 2023 [DOI](#); Steib et al., 2022 [DOI](#)). Alternatively, imaging techniques that allow super-resolution like DNA-PAINT, could help to achieve higher resolution (sub 5 nm resolution) (Agasti et al., 2017 [DOI](#); Schueder et al., 2023 [DOI](#)). Thirdly, apart from their small size, most *Drosophila* neuronal nuclei are found in dense regions, with significant overlap in 2D images. Thus, 3D imaging could be used to improve nuclei segmentation (Pachitariu & Stringer, 2022 [DOI](#)), but still leaves open the problem of overlapping neuronal projections.

Therefore, we explored several computational techniques to integrate SRT with single-cell RNA-seq, including grid-rasterization, nuclei-segmentation and neighborhood embedding. Given the dense packing of the small nuclei in the fly brain, and the likely presence of mRNA transport away from the nucleus of origin, nuclei-segmentation leads to a loss of information compared to the other nuclei-unaware techniques. We further found that the mapping of some clusters is dominated by the presence of strong marker genes. Therefore, we predict strong improvements when using larger gene panels. Currently, sequencing-based methods such as Stereo-seq hold an advantage in number of genes that can be profiled (Wang et al., 2022 [DOI](#)), but smFISH-based approaches are scaling rapidly and will soon allow to routinely image 1000s of genes at higher resolutions (Fang et al., 2022 [DOI](#)).

The most exciting advantage of SRT compared to single-cell techniques applied to cells or nuclei in suspension, is the possibility to study mRNA localization within the tissue-context. It was previously reported in the embryo that large numbers of mRNAs are non-homogeneously distributed and thus likely transported (Jambor et al., 2015 [DOI](#); Lécuyer et al., 2007 [DOI](#); Tomancak et al., 2002 [DOI](#)). In our SRT data, we found localized mRNAs in at least three different cell types. First, we found that different trypsin isoforms locate to opposite apical-basal locations in enterocytes of the adult gut. Such apical-basal mRNA transport processes are well described in the embryonic epidermis (Bullock et al., 2006 [DOI](#)), however, to our knowledge not yet known in the adult fly gut epidermis, where secretion of the enzymes at the correct side of the cell should be critical.

Second, in the brain, we identified localized patterns of mRNAs coding for the neuropeptides Pdf and Ilp2. These mRNAs are likely transported into axons or dendrites, away from the cell body, a process often critical for brain development and function (Holt & Bullock, 2009 [DOI](#)). As both genes

are important regulators of adult physiology, the identified localization patterns may inspire future studies.

Third, in muscle cells, SRT revealed different patterns of mRNA distributions within muscle fibers that suggest both mRNA transport and local synthesis in a subset of nuclei. Most prominently, *sIs* mRNA remains close to the nuclei, whereas some muscle regions display alternating striped patterns of *Act88F* and *TpnC4* mRNAs that recapitulate the functional specializations within the muscle fibers, with myofibrils, mitochondria and ER being present at defined locations (Luis & Schnorrer, 2021 [↗](#)). It has been proposed that ribosomes are concentrated at particular locations in mammalian muscles to build translational hotspots (Denes et al., 2021 [↗](#); Lewis et al., 2018 [↗](#); Rudolph et al., 2019 [↗](#)). Whether such concentrations exist in fly muscles is not known, and it will require future investigations to identify the localization mechanisms and the functional significance of localized mRNAs concentrations. Finally, we propose the existence of heterogeneous nuclei in the IFMs, with terminal nuclei facing the muscle-tendon attachment sites showing an enrichment of *sIs* mRNA and thus likely transcribing more *sIs* mRNA. *SIs* protein is the titin homolog, a critical component of the sarcomeric Z-disks (Loreau et al., 2023 [↗](#)), also located at the terminal Z-disk linking each force-probing myofibril to the tendon cells (Green et al., 2018 [↗](#)). Similarly, in mammalian muscles, it was recently found that the terminal muscle fiber nuclei, which face the tendon cells, apply a specialized transcriptional program (Kim et al., 2020 [↗](#); Esteves de Lima et al., 2021 [↗](#)). Interestingly, the FCA data of cross-tissue muscle cells did not reveal transcriptionally heterogeneous subclusters of nuclei, showing the benefit of more sensitive spatial techniques (Li et al., 2022 [↗](#)).

To conclude, we have shown that medium throughput SRT on entire fly head and body samples is methodologically possible and leads to informative new discoveries that will spark interesting follow-up studies. All our data can be freely explored on <https://spatialfly.aertslab.org> [↗](#). This proof-of-concept study opens avenues for potential full 3D slicing and imaging of an adult fly in its entirety at high resolution.

Methods

Full fly sectioning for MC

To ensure full adult development, adult male Luminy flies were isolated and left for 3 days at 25°C. After this step, the flies were put on ice for 15 minutes, and their wings were clipped. Flies were then transferred to a freezing mold alive, embedded in optimal cutting temperature compound (OCT compound, VWR), and frozen in liquid nitrogen. Flies were sectioned with a cryostat (Leica CM 3050 S, Leica Biosystems, Germany). 10 µm sections were transferred to coverslips coated with gelatin (porcine skin, 300g Bloom Type A, Merck) and stored at -80°C. 5 body sections from 1 male were imaged with MC.

Fly head sectioning for MC

Adult flies (male and females) were anesthetized on a fly pad using CO₂ gas, to cut off their heads with a scalpel. Heads were then placed on a pre-cooled metal surface on dry ice and covered with a drop of OCT. Frozen OCT blocks were stored at -80°C until sectioning. Head sections of 10 µm thickness were produced with a Leica CM3050 S cryostat and placed on uncoated coverslips, which were stored at -80°C. 13 head sections from 12 different flies were imaged with MC.

Molecular Cartography (MC)

a. Gene selection and probe design

Guided by the fly cell atlas, 150 genes were selected to be the most informative for spatial transcriptomics, 50 genes for the whole-body samples (Table S1) and 100 genes for the head samples (Table S2). These genes were selected based on different criteria including: marking specific cell populations, strong single-cell co-expression, marking uncharacterized cell populations or showing broad expression.

The probes for the selected genes were designed using Resolve Biosciences' proprietary design algorithm. To speed up the process, the calculation of computationally expensive parts, especially the off-target searches, the selection of probe sequences was not performed randomly, but limited to sequences with high success rates. To filter highly repetitive regions, the abundance of k-mers was obtained from the background transcriptome using Jellyfish (Marçais & Kingsford, 2011 [↗](#)). Every target sequence was scanned once for all k-mers and those regions with rare k-mers were preferred as seeds for full probe design. A probe candidate was generated by extending a seed sequence until a certain target stability was reached. A set of simple rules was applied to discard sequences that were found experimentally to cause problems. After these fast screens, every kept probe candidate was mapped to the background transcriptome using TheronucleotideBLAST (Gans & Wolinsky, 2008 [↗](#)) and probes with stable off-target hits were discarded. Specific probes were then scored based on the number of on-target matches (isoforms), which were weighted by their associated APPRIS level (Rodriguez et al., 2018 [↗](#)), favoring principal isoforms over others. A bonus was added if the binding site was inside the protein-coding region. From the pool of accepted probes, the final set was composed by greedily picking the highest scoring probes. Catalog numbers for the specific probes are available upon request at Resolve Biosciences.

b. Molecular Cartography

Samples were then sent to Resolve Biosciences on dry ice for analysis. Upon arrival, tissue sections were thawed and rehydrated with isopropanol, followed by 1 min washes in 95% ethanol and then 70% ethanol at room temperature. The samples were used for Molecular Cartography (100-plex combinatorial single molecule fluorescence in situ hybridization) according to the manufacturer's instructions (protocol v.1.3; available for registered users), starting with the aspiration of ethanol and the addition of buffer DST1 followed by tissue priming and hybridization. Briefly, tissues were primed for 30 min at 37 °C followed by overnight hybridization of all probes specific for the target genes (see below for probe design details and target list). Samples were washed the next day to remove excess probes and fluorescently tagged in a two-step color development process. Regions of interest were imaged as described below and fluorescent signals were removed during decolorization. Color development, imaging and decolorization were repeated for multiple cycles to build a unique combinatorial code for every target gene that was derived from raw images as described below.

c. Imaging

Samples were imaged on a Zeiss Celldiscoverer 7, using the ×50 Plan Apochromat water immersion objective with an NA of 1.2 and the ×0.5 magnification changer, resulting in a ×25 final magnification. Standard CD7 LED excitation light source, filters and dichroic mirrors were used together with customized emission filters optimized for detecting specific signals. Excitation time per image was 1,000 ms for each channel (4,6-diamidino-2-phenylindole (DAPI) was 20 ms). A z-stack was taken at each region with a distance per z-slice according to the Nyquist–Shannon sampling theorem. The custom CD7 CMOS camera (Zeiss AxioCam Mono 712) was used. For each region, a z-stack per fluorescent color (two colors) was imaged per imaging round. A total of eight imaging rounds were conducted for each position, resulting in 16 z-stacks per region. The

completely automated imaging process per round (including water immersion generation and precise relocation of regions to image in all three dimensions) was realized by a custom Python script using the scripting API of the Zeiss ZEN software (open application development).

d. Spot segmentation

The algorithms for spot segmentation were written in Java and are based on the ImageJ library functionalities. The iterative closest point algorithm is written in C++ based on the libpointmatcher library (<https://github.com/ethz-asl/libpointmatcher> .

e. Preprocessing

As a first step, all images were corrected for background fluorescence. A target value for the allowed number of maxima was determined based upon the area of the slice in μm^2 multiplied by the factor 0.5. This factor was empirically optimized. The brightest maxima per plane were determined, based upon an empirically optimized threshold. The number and location of the respective maxima was stored. This procedure was conducted for every image slice independently. Maxima that did not have a neighboring maximum in an adjacent slice (called a z group) were excluded. The resulting maxima list was further filtered in an iterative loop by adjusting the allowed thresholds for (Babs - Bback) and (Bperi - Bback) to reach a feature target value (Babs: absolute brightness; Bback: local background; Bperi: background of periphery within one pixel). These feature target values were based upon the volume of the three-dimensional (3D) image. Only maxima still in a z-group of at least two after filtering were passing the filter step. Each z-group was counted as one hit. The members of the z-groups with the highest absolute brightness were used as features and written to a file. They resemble a 3D point cloud.

f. Final signal segmentation and decoding

To align the raw data images from different imaging rounds, images had to be corrected. To do so, the extracted feature point clouds were used to find the transformation matrices. For this purpose, an iterative closest point cloud algorithm was used to minimize the error between two-point clouds. The point clouds of each round were aligned to the point cloud of round one (the reference point cloud). The corresponding point clouds were stored for downstream processes. Based upon the transformation matrices, the corresponding images were processed by a rigid transformation using trilinear interpolation. The aligned images were used to create a profile for each pixel consisting of 16 values (16 images from two color channels in eight imaging rounds). The pixel profiles were filtered for variance from zero normalized by total brightness of all pixels in the profile. Matched pixel profiles with the highest score were assigned as an ID to the pixel. Pixels with neighbors having the same ID were grouped. The pixel groups were filtered by group size, number of direct adjacent pixels in group and number of dimensions with size of two pixels. The local 3D maxima of the groups were determined as potential final transcript locations. Maxima were filtered by number of maxima in the raw data images where a maximum was expected. Remaining maxima were further evaluated by the fit to the corresponding code. The remaining maxima were written to the results file and considered to resemble transcripts of the corresponding gene. The ratio of signals matching to codes used in the experiment and signals matching to codes not used in the experiment were used as estimation for specificity (false positives).

From Resolve Biosciences, the authors received the raw DAPI signal containing tiff image files, with gene localization count tables.

g. Gene visualization

Genes were plotted using Python scripts. Marker sizes were scaled by gene density to increase visibility of patterns.

Hybridization chain reaction on fly thoraces

a. Thorax preparation for HCR-FISH

5-day old Luminy males, raised at 25°C, were put on ice for 15 min before clipping their wings, head and abdomen. Thoraces were then transferred in PAXgene fixative (Resolve Biosciences) for 1 hour at room temperature, followed by 2 hours in stabilization buffer (Resolve Biosciences) at room temperature and overnight in 30% sucrose in 1x PBS at 4°C. The next day, thoraces were transferred to OCT (VWR) and immediately frozen in liquid nitrogen. Fly thoraces were sectioned with a cryostat (Leica CM 3050 S, Leica Biosystems, Germany). 16 µm sections were transferred to slides coated with gelatin (porcine skin, 300g Bloom Type A, Merck)

b. Probe design

Pairs of DNA 25-mer oligos were designed to hybridize on *Act88F*, *TpnC4*, *sls* and *Mhc* transcripts. Except for *TpnC4*, for which we designed 12 pairs of oligos, 20 pairs were designed for each mRNA, following the principles presented by (Choi et al., 2018 [DOI](#)). Sequences are provided in the Table S4.

c. Hybridization chain reaction protocol

We followed the plated-cells protocol presented in (Choi et al., 2018 [DOI](#)) with slight modifications to apply it on fly sections.

Day 1: wash 2x with PBS, add ice cold 70% ethanol and incubate overnight at -20°C.

Day 2: aspirate ethanol, wash 2x with 2x SSC (Invitrogen), incubate in 300 µL of “30% probe hybridization buffer” (Molecular Instruments) for 30 min at 37°C. Incubate with 1.2 pmol of each probe mixture to 300 µL of “30% probe hybridization buffer” at 37°C overnight.

Day 3: wash 4x 5 min with 300 µL of “30% probe wash buffer” (Molecular Instruments) at 37°C, wash samples 2x 5 min with 5x SSCT (0.1% Tween 20 in 5x SSC) at room temperature. Incubate samples in 300 µL of amplification buffer (Molecular Instruments) for 30 min at room temperature. Prepare 18 pmol of each fluorescently labeled hairpin (HCR Amplifiers B1-488 for *Act88F*, B3-546 for *TpnC4*, and B5-647 for *SlS* and *Mhc*, Molecular Instruments) by snap cooling (heat at 95°C for 90 seconds and cool to room temperature in a dark drawer for 30 min). Prepare hairpin solution by adding all snap-cooled hairpins to 300 µL of amplification buffer at room temperature. Incubate samples overnight (12 - 16 h) in the dark at room temperature.

Day 4: wash 1x with 300 µL of 5x SSCT 5 min, stain with DAPI for 30 min and wash for 5 min with 300 µL of 5x SSCT at room temperature and mount in glycerol DABCO.

d. Imaging of fly thoraces

Thoraces were imaged with a point scanning confocal microscope (Leica SP8) using a 20x 0.75 NA glycerol immersion objective (Leica HC CS2 Plan Apo 20x 0.75 NA Imm).

Clustering analysis of mRNA species based on proximity

Based on the observation that groups of mRNA species appeared to cluster into separated regions of sections, we devised a simple method to automatically extract these regions based on the proximity of different mRNA. mRNA species that were identified as being close were then gathered into the same cluster.

In practice, we first computed the proximity of mRNA species by pair (see Figure 2 – figure supplement 3). Each localization of two mRNA species was transformed into disks of fixed diameters, each disk being centered on a given mRNA localization; the diameter used here was 4 μm . To generate one surface for a species and avoid counting multiple times the same area, disks of a given mRNA species were merged if they were overlapping. We then computed their overlap surface from the surfaces obtained from two different mRNA species. The proximity of one mRNA species (mRNA1) versus the over one (mRNA2) was defined as the ratio between the overlap surface and surface of the second mRNA species (mRNA2):

$$\text{Proximity mRNA}_{1 \text{ in } 2} = \frac{\text{Overlap mRNA}_{1 \text{ and } 2}}{\text{Surface mRNA}_2}$$

Reciprocally,

$$\text{Proximity mRNA}_{2 \text{ in } 1} = \frac{\text{Overlap mRNA}_{1 \text{ and } 2}}{\text{Surface mRNA}_1}$$

The calculation of proximity then allowed us to define a distance between two mRNA species:

$$\text{Distance mRNA}_{1 \text{ and } 2} = 2 - (\text{Proximity mRNA}_{1 \text{ in } 2} + \text{Proximity mRNA}_{2 \text{ in } 1})$$

mRNA species that show perfect overlap get a distance of 0 in this metric, whereas mRNA species that do not show any overlap would get a distance of 2.

Finally, this metric was used in a hierarchical clustering analysis using Ward's method. Clusters were then extracted from this analysis.

5 μm x 5 μm grid analysis

a. Quantification

Samples were rasterized in a square grid of 36 by 36 pixels (1 pixel = 0.138 μm , 36 pixels = 4.968 μm). All counts within this area were summed up. This led to a square by gene matrix, with for every square the mean x and y spatial coordinates of the square's dimensions.

b. SCANPY body

All body samples (5) were analyzed together in SCANPY (1.9.3) (Wolf et al., 2018). An increment of 1000 was added to both spatial x- and y-coordinates to arrange all samples together. Only squares with more than 3 counts were kept, leading to 132,642 squares with information for 50 genes. The data was subsequently normalized with 10,000 as size factor and log transformed. This matrix was then used as input for PCA, after which 40 components were retained by evaluating variance ratio plots. Harmony was then used to correct for batch effects between samples. 30 PCs were used to calculate neighbors, Leiden clustering (resolutions 0.2, 0.5, 1 and 2) and UMAP embeddings.

c. SCANPY head

All head samples (13) were analyzed together in SCANPY (1.9.3) (Wolf et al., 2018). An increment of 1000 was added to both spatial x- and y-coordinates to arrange all samples together. Only squares with more than 3 counts were kept, leading to 83,064 squares with information for 99 genes (*ninaE* was discarded from most samples due to optical crowding). The data was subsequently normalized with 10,000 as size factor and log transformed. This matrix was then used as input for PCA, after which 30 components were retained by evaluating variance ratio plots. Harmony was then used to correct for batch effects between samples. 25 PCs were used to calculate neighbors, Leiden clustering (resolutions 0.2, 0.5, 1 and 2) and UMAP embeddings.

d. Head OL vs CB vs PR differential expression

Leiden resolution 1 was used to create average gene expression profiles for clusters. Clusters with mean expression of para > 0.05 were selected as neuronal clusters. These were subsequently manually assigned to either Photoreceptor (PR), Optic lobe (OL) or Central brain (CB) regions based on location. Next, the `rank_genes_groups` function from SCANPY was used to calculate differential genes for the regions based on a t-test.

e. Gene set enrichment

In the body, we selected marker genes that were assigned to several categories. We then summed the expression of genes belonging to the same category to derive gene set signatures. The following gene sets were used: muscle (*Mhc*, *sls*, *CG32121*), neurons (*elav*, *Syt1*, *Sh*, *acj6*, *ey*, *VAcHT*, *Gad1*, *VGlut*, *nAChRalpha7*), glia (*repo*, *alrm*), epithelia (pure: *grh*, *hth*, gut: *alphaTry*, *betaTry*), heart (*tin*, *Hand*), fat body (*AkhR*, *FASN2*), oenocyte (*FASN2*), male reproductive system (*Awh*, *eyg*, *svp*) and hemocytes (*Hml*). Spots in the grid were assigned to a category based on Z-scores. If z-normalized expression > 1, the spot was assigned to the category. When conflicts arose, the following hierarchy was used: muscle > epithelia > glia > neurons > male reproductive system > fat body > oenocyte > hemocyte > heart.

f. Lasso matching

We used lasso regression as implemented in sklearn (1.2.2). Averaged gene expression profiles of single-cell clusters were matched with expression profiles of the grid-based squares. The regression model was run using Lasso (alpha = 1, positive = True), forcing all coefficients to be positive, with all genes. Higher weights for the single-cell clusters corresponds to a higher similarity between the cluster and the square. We only used weights > 0.2 as confident matches. Spots were then assigned to the cluster based on the highest weight.

Nuclei segmentation

Nuclei in DAPI images were segmented with Cellprofiler (version 4.0.7) (Stirling et al., 2021 [link](#)). Features of the DAPI images were enhanced (*EnhanceOrSuppressFeatures*, *Enhance*, *Speckles*) using a feature size of 100 pixels. Initial nuclei were detected using a size of 10- 100 pixels, global Otsu thresholding with two classes, intensity for distinguishing clumped objects, *propagate* for drawing dividing lines. Initial nuclei were extended by 25 pixels. We then collected all transcripts inside a segmented cell to compute counts of each gene per segmented cell.

Tangram

Tangram (version 1.0.2) (Biancalani et al., 2021 [link](#)) was used to project cell type labels from single cell data. Here we used the *cell* mapping mode to map single cell data to each spatial slice separately. The mapping was computed on a NVIDIA A100-SXM4-80GB GPU. Prior to computing the mapping, segmented cells with less than three expressed genes and genes that were expressed in less than three cells were removed. Single cells and segmented cells were log-normalized in SCANPY (Wolf et al., 2018 [link](#)) with a target-sum of 10^4 . All genes that are shared between the spatial data and single cell data after filtering were used for the mapping. Cell type labels for fly head samples were assigned separately for glia, optic lobe (OL), central brain (CB) and unknown clusters (UNK) by considering only the subset cell types that we grouped into the category. A cell type label was assigned to a segmented cell based on the 95%-quantile of the mapping scores for a cell type (unless stated otherwise) if this resulted in a unique label assignment. Cell type labels for the grid rasterized spatial data were assigned in the same manner as we did for the segmented cells. The only exception being, that we down-sampled the number of cells to a maximum of 5000 cells per cell type in the single cell data to limit the amounts of required video memory.

Neighborhood embedding and SpaGE

Segmentation-free analysis of fly head and body SRT datasets have been carried out using `space2vec` (Partel & Wählby, 2021 [↗](#)). Briefly, spatial graphs of all mRNAs are first constructed for each different sample. Then, a graph convolution neural network trained in a self-supervised fashion projects each mRNA into an embedding space based on its spatial neighborhood composition. Thus, mRNAs that share similar neighbors are mapped close together in the embedding space. Downstream clustering or visualization of mRNA embeddings unveil spatial gene expression patterns described by specific combinations of genes at subcellular resolution. Pseudo-cell counts have been generated for each mRNA by aggregating counts of k neighboring mRNAs (i.e. $k=100$) in the embedding space. Finally, integration of reference scRNA-seq datasets with spatial pseudo-cell counts have been implemented by projecting both datasets into a common shared space using SpaGE (Abdelaal et al., 2020 [↗](#)). Thereafter, cell type labels have been transferred from scRNA-seq cells to spatial pseudo-cells by kNN imputation (with $k=100$).

mRNA localization in muscle - distance to nuclei

For every detected mRNA molecule, the distance to the closest nucleus was calculated. First, we calculated a mask to segment the indirect flight muscle using *Act88F* mRNA spots. Using `opencv2`, we performed a Gaussian blur ($ksize = 5 \times 5$, $\sigma = 1$), followed by two erosion steps (5×5 and 4×4) to remove sparse signals. Then, the spots were dilated thrice (50×50), followed by a final Gaussian blur ($ksize = 5 \times 5$, $\sigma = 1$). Only other mRNA species that overlap with this mask were kept for the distance analysis.

The nucleus segmentation was loaded as a black-white image. The `nonzero` function from `opencv2` was used to find segmented pixels. Then for each mRNA spot, Euclidean distances were calculated to the segmented pixels, after which they were assigned to the closest pixel. To optimize the calculations, `joblib`'s `Parallel` function was used. This was repeated over all body samples and results were combined. Only genes for which at least 100 mRNA spots were detected across samples and not localized at the edges (faulty segmentation along the muscle boundary) were kept: *Act88F*, *Mhc*, *TpnC4*, *sls*, *salm* and *CG32121*.

Data Availability

All data (raw DAPI .tiff files, mRNA localization tables) are available on <https://spatialfly.aertslab.org/> [↗](#). Supplementary Table S1 and S2 list the genes used as probes. Probe design was performed by Resolve Biosciences. Gene panels are commercially available for use. HCR probes are provided in Table S4.

Acknowledgements

We thank VIB Tech Watch, Resolve Biosciences, and Jeroen Aerts for facilitating early access to the Molecular Cartography platform. Computing was performed on VSC (Vlaams Supercomputer Centrum). We thank Bruno C. Vellutini for the help with guiding us through Hybridization Chain Reaction preparation and protocol. We thank Pavel Tomancak for hosting P.M. and providing support to perform HCR, and the MPI-CBG Light Microscopy Facility and Computer Services Facility for support.

Additional information

Funding

This work was in part funded by ERC AdG Genome2Cells (S.A.; 101054387); a FWO PhD Fellowship to (J.J.; 1199518N); a FWO Postdoctoral fellowship (N.H.; 1273822N); GP was funded by Opening The Future. This work was supported by the Centre National de la Recherche Scientifique (CNRS, F.S.), Aix-Marseille University (P.M.), the European Research Council under the European Union's Horizon 2020 Programme (ERC-2019-SyG 856118 F.S.), FP/2007-2013)/ERC Grant 310939 (F.S.) and by funding from France 2030, the French Government program managed by the French National Research Agency (ANR-16-CONV- 0001) and from Excellence Initiative of Aix-Marseille University - A*MIDEX (Turing Centre for Living Systems).

Code Availability

Code for the website (<https://spatialfly.aertslab.org/>) is available on Github (https://github.com/aertslab/spatial_fly_website). In this folder scripts to perform grid, segmentation and neighborhood-embedding analysis can also be found. A pipeline can be found here: <https://github.com/aertslab/SpatialNF>.

Contributions

S.A., F.S. and J.J. conceived the study. K.S. and J.I. prepared head samples. P.M. prepared body samples. J.J., P.M., N.H. and G.P. performed computational analyses. J.J. performed grid-based rasterization. P.M. performed mRNA overlap analysis. N.H. performed nuclei segmentation and Tangram label transfer. G.P. performed neighborhood embedding analysis. F.S., P.M. and J.J. performed muscle analysis. J.J. created the website with assistance of G.H. N.H. and G.P. developed the spatial analysis pipeline. J.J., F.S., S.A. and P.M. wrote and revised the manuscript.

Additional files

Supplementary Figures and Legend

Table S1

Table S2

Table S3

Table S4

References

- Abdelaal T., Mourragui S., Mahfouz A., Reinders M. J. T 2020) **SpaGE: Spatial Gene Enhancement using scRNA-seq** *Nucleic Acids Research* **48**:e107 <https://doi.org/10.1093/nar/gkaa740>
- Agasti S. S., Wang Y., Schueder F., Sukumar A., Jungmann R., Yin P 2017) **DNA- barcoded labeling probes for highly multiplexed Exchange-PAINT imaging** *Chemical Science* **8**:3080–3091 <https://doi.org/10.1039/C6SC05420J>
- Agianian B., Krzic U., Qiu F., Linke W. A., Leonard K., Bullard B 2004) **A troponin switch that regulates muscle contraction by stretch instead of calcium** *The EMBO Journal* **23**:772–779 <https://doi.org/10.1038/sj.emboj.7600097>
- Avellaneda J., Rodier C., Daian F., Brouilly N., Rival T., Luis N. M., Schnorrer F 2021) **Myofibril and mitochondria morphogenesis are coordinated by a mechanical feedback mechanism in muscle** *Nature Communications* **12**:Article 1 <https://doi.org/10.1038/s41467-021-22058-7>
- Benton R 2022) **Drosophila olfaction: Past, present and future** *Proceedings of the Royal Society B: Biological Sciences* **289**:20222054 <https://doi.org/10.1098/rspb.2022.2054>
- Biancalani T., Scalia G., Buffoni L., Avasthi R., Lu Z., Sanger A., Tokcan N., Vanderburg C. R., Segerstolpe Å., Zhang M., Avraham-Davidi I., Vickovic S., Nitzan M., Ma S., Subramanian A., Lipinski M., Buenrostro J., Brown N. B., Fanelli D., ..., Regev A 2021) **Deep learning and alignment of spatially resolved single-cell transcriptomes with Tangram** *Nature Methods* **18**:Article 11 <https://doi.org/10.1038/s41592-021-01264-7>
- Bruusgaard J. C., Liestøl K., Ekmark M., Kollstad K., Gundersen K 2003) **Number and spatial distribution of nuclei in the muscle fibres of normal mice studied in vivo** *The Journal of Physiology* **551**:467–478 <https://doi.org/10.1113/jphysiol.2003.045328>
- Bullock S. L., Nicol A., Gross S. P., Zicha D 2006) **Guidance of bidirectional motor complexes by mRNA cargoes through control of dynein number and activity** *Current Biology: CB* **16**:1447–1452 <https://doi.org/10.1016/j.cub.2006.05.055>
- Chaturvedi D., Reichert H., Gunage R. D., VijayRaghavan K 2017) **Identification and functional characterization of muscle satellite cells in Drosophila** *eLife* **6**:e30107 <https://doi.org/10.7554/eLife.30107>
- Chen K. H., Boettiger A. N., Moffitt J. R., Wang S., Zhuang X 2015) **Spatially resolved, highly multiplexed RNA profiling in single cells** *Science* **348**:aaa6090 <https://doi.org/10.1126/science.aaa6090>
- Choi H. M. T., Schwarzkopf M., Fornace M. E., Acharya A., Artavanis G., Stegmaier J., Cunha A., Pierce N. A 2018) **Third-generation in situ hybridization chain reaction: Multiplexed, quantitative, sensitive, versatile, robust** *Development* **145**:dev165753 <https://doi.org/10.1242/dev.165753>
- Das S., Vera M., Gandin V., Singer R. H., Tutucci E 2021) **Intracellular mRNA transport and localized translation** *Nature Reviews Molecular Cell Biology* **22**:483–504 <https://doi.org/10.1038>

/s41580-021-00356-8

Davie K., Janssens J., Koldere D., De Waegeneer M., Pech U., Kreft L., Aibar S., Makhzami S., Christiaens V., Bravo González-Blas C., Poovathingal S., Hulselmans G., Spanier K. I., Moerman T., Vanspauwen B., Geurs S., Voet T., Lammertyn J., Thienpont B., ..., Aerts S. 2018) **A Single-Cell Transcriptome Atlas of the Aging Drosophila Brain** *Cell* **174**:982–998 <https://doi.org/10.1016/j.cell.2018.05.057>

Denes L. T., Kelley C. P., Wang E. T. 2021) **Microtubule-based transport is essential to distribute RNA and nascent protein in skeletal muscle** *Nature Communications* **12**:Article 1 <https://doi.org/10.1038/s41467-021-26383-9>

Díaz M. M., Schlichting M., Abruzzi K. C., Long X., Rosbash M. 2019) **Allatostatin- C/AstC-R2 Is a Novel Pathway to Modulate the Circadian Activity Pattern in Drosophila** *Current Biology* **29**:13–22 <https://doi.org/10.1016/j.cub.2018.11.005>

Dix D. J., Eisenberg B. R. 1990) **Myosin mRNA accumulation and myofibrillogenesis at the myotendinous junction of stretched muscle fibers** *The Journal of Cell Biology* **111**:1885–1894 <https://doi.org/10.1083/jcb.111.5.1885>

Dorkenwald S., Matsliah A., Sterling A. R., Schlegel P., Yu S., McKellar C. E., Lin A., Costa M., Eichler K., Yin Y., Silversmith W., Schneider-Mizell C., Jordan C. S., Brittain D., Halageri A., Kuehner K., Ogedengbe O., Morey R., Gager J., ..., Zandawala M. 2024) **Neuronal wiring diagram of an adult brain** *Nature* **634**:124–138 <https://doi.org/10.1038/s41586-024-07558-y>

Eng C.-H. L., Lawson M., Zhu Q., Dries R., Koulana N., Takei Y., Yun J., Cronin C., Karp C., Yuan G.-C., Cai L. 2019) **Transcriptome-scale super-resolved imaging in tissues by RNA seqFISH+** *Nature* **568**:Article 7751 <https://doi.org/10.1038/s41586-019-1049-y>

Esteves de Lima J., Blavet C., Bonnin M.-A., Hirsinger E., Comai G., Yvernogeu L., Delfini M.-C., Bellenger L., Mella S., Nassari S., Robin C., Schweitzer R., Fournier-Thibault C., Jaffredo T., Tajbakhsh S., Relaix F., Duprez D. 2021) **Unexpected contribution of fibroblasts to muscle lineage as a mechanism for limb muscle patterning** *Nature Communications* **12**:Article 1 <https://doi.org/10.1038/s41467-021-24157-x>

Fan Y., Andrusivová Ž., Wu Y., Chai C., Larsson L., He M., Luo L., Lundeberg J., Wang B. 2023) **Expansion spatial transcriptomics** *Nature Methods* **20**:Article 8 <https://doi.org/10.1038/s41592-023-01911-1>

Fang R., Xia C., Close J. L., Zhang M., He J., Huang Z., Halpern A. R., Long B., Miller J. A., Lein E. S., Zhuang X. 2022) **Conservation and divergence of cortical cell organization in human and mouse revealed by MERFISH** *Science* **377**:56–62 <https://doi.org/10.1126/science.abm1741>

Folker E., Baylies M. 2013) **Nuclear positioning in muscle development and disease** *Frontiers in Physiology* **4** <https://www.frontiersin.org/articles/10.3389/fphys.2013.00363>

Fyrberg E. A., Mahaffey J. W., Bond B. J., Davidson N. 1983) **Transcripts of the six Drosophila actin genes accumulate in a stage- and tissue-specific manner** *Cell* **33**:115–123 [https://doi.org/10.1016/0092-8674\(83\)90340-9](https://doi.org/10.1016/0092-8674(83)90340-9)

Gans J. D., Wolinsky M. 2008) **Improved assay-dependent searching of nucleic acid sequence databases** *Nucleic Acids Research* **36**:e74 <https://doi.org/10.1093/nar/gkn301>

- Green H. J., Griffiths A. G., Ylänne J., Brown N. H (2018) **Novel functions for integrin-associated proteins revealed by analysis of myofibril attachment in *Drosophila*** *eLife* **7**:e35783 <https://doi.org/10.7554/eLife.35783>
- Grenier T., Leulier F (2020) **How commensal microbes shape the physiology of *Drosophila melanogaster*** *Current Opinion in Insect Science* **41**:92–99 <https://doi.org/10.1016/j.cois.2020.08.002>
- Groiss S., Pabst D., Faber C., Meier A., Bogdoll A., Unger C., Nilges B., Strauss S., Förderl-Höbenreich E., Hardt M., Geipel A., Reinecke F., Korfhage C., Zatloukal K (2021) **Highly resolved spatial transcriptomics for detection of rare events in cells** *bioRxiv* <https://doi.org/10.1101/2021.10.11.463936>
- Han X., Wang M., Liu C., Trush O., Takayama R., Akiyama T., Naito T., Tomomizu T., Imamura K., Sato M (2020) **DWnt4 and DWnt10 Regulate Morphogenesis and Arrangement of Columnar Units via Fz2/PCP Signaling in the *Drosophila* Brain** *Cell Reports* **33**:108305 <https://doi.org/10.1016/j.celrep.2020.108305>
- Holt C. E., Bullock S. L (2009) **Subcellular mRNA Localization in Animal Cells and Why It Matters** *Science* **326**:1212–1216 <https://doi.org/10.1126/science.1176488>
- Jambor H., Surendranath V., Kalinka A. T., Mejstrik P., Saalfeld S., Tomancak P (2015) **Systematic imaging reveals features and changing localization of mRNAs in *Drosophila* development** *eLife* **4**:e05003 <https://doi.org/10.7554/eLife.05003>
- Jiang H., Edgar B. A (2011) **Intestinal stem cells in the adult *Drosophila* midgut** *Experimental Cell Research* **317**:2780–2788 <https://doi.org/10.1016/j.yexcr.2011.07.020>
- Kaya-Çopur A., Marchiano F., Hein M. Y., Alpern D., Russeil J., Luis N. M., Mann M., Deplancke B., Habermann B. H., Schnorrer F (2021) **The Hippo pathway controls myofibril assembly and muscle fiber growth by regulating sarcomeric gene expression** *eLife* **10**:e63726 <https://doi.org/10.7554/eLife.63726>
- Kim M., Franke V., Brandt B., Lowenstein E. D., Schöwel V., Spuler S., Akalin A., Birchmeier C (2020) **Single-nucleus transcriptomics reveals functional compartmentalization in syncytial skeletal muscle cells** *Nature Communications* **11**:Article 1 <https://doi.org/10.1038/s41467-020-20064-9>
- Klein A. M., Mazutis L., Akartuna I., Tallapragada N., Veres A., Li V., Peshkin L., Weitz D. A., Kirschner M. W (2015) **Droplet Barcoding for Single-Cell Transcriptomics Applied to Embryonic Stem Cells** *Cell* **161**:1187–1201 <https://doi.org/10.1016/j.cell.2015.04.044>
- Lécuyer E., Yoshida H., Parthasarathy N., Alm C., Babak T., Cerovina T., Hughes T. R., Tomancak P., Krause H. M (2007) **Global Analysis of mRNA Localization Reveals a Prominent Role in Organizing Cellular Architecture and Function** *Cell* **131**:174–187 <https://doi.org/10.1016/j.cell.2007.08.003>
- Lewis Y. E., Moskovitz A., Mutlak M., Heineke J., Caspi L. H., Kehat I (2018) **Localization of transcripts, translation, and degradation for spatiotemporal sarcomere maintenance** *Journal of Molecular and Cellular Cardiology* **116**:16–28 <https://doi.org/10.1016/j.yjmcc.2018.01.012>
- Li H., Janssens J., De Waegeneer M., Kolluru S. S., Davie K., Gardeux V., Saelens W., David F. P. A., Brbić M., Spanier K., Leskovec J., McLaughlin C. N., Xie Q., Jones R. C., Brueckner K., Shim J.,

- Tattikota S. G., Schnorrer F., Rust K., ..., Zinzen R. P. 2022) **Fly Cell Atlas: A single-nucleus transcriptomic atlas of the adult fruit fly** *Science* **375**:eabk2432 <https://doi.org/10.1126/science.abk2432>
- Loreau V., Rees R., Chan E. H., Taxer W., Gregor K., Mußil B., Pitaval C., Luis N. M., Mangeol P., Schnorrer F., Görlich D 2023) **A nanobody toolbox to investigate localisation and dynamics of Drosophila titins and other key sarcomeric proteins** *eLife* **12**:e79343 <https://doi.org/10.7554/eLife.79343>
- Lu T.-C., Brbić M., Park Y.-J., Jackson T., Chen J., Kolluru S. S., Qi Y., Katheder N. S., Cai X. T., Lee S., Chen Y.-C., Auld N., Liang C.-Y., Ding S. H., Welsch D., D'Souza S., Pisco A. O., Jones R. C., Leskovec J., ..., Li H 2023) **Aging Fly Cell Atlas identifies exhaustive aging features at cellular resolution** *Science* **380**:eadg0934 <https://doi.org/10.1126/science.adg0934>
- Lubeck E., Coskun A. F., Zhiyentayev T., Ahmad M., Cai L 2014) **Single-cell in situ RNA profiling by sequential hybridization** *Nature Methods* **11**:Article 4 <https://doi.org/10.1038/nmeth.2892>
- Luis N. M., Schnorrer F 2021) **Mechanobiology of muscle and myofibril morphogenesis** *Cells & Development* **168**:203760 <https://doi.org/10.1016/j.cdev.2021.203760>
- Marçais G., Kingsford C 2011) **A fast, lock-free approach for efficient parallel counting of occurrences of k-mers** *Bioinformatics* **27**:764–770 <https://doi.org/10.1093/bioinformatics/btr011>
- Matsubayashi Y., Louani A., Dragu A., Sánchez-Sánchez B. J., Serna-Morales E., Yolland L., Gyoergy A., Vizcay G., Fleck R. A., Heddleston J. M., Chew T.-L., Siekhaus D. E., Stramer B. M 2017) **A Moving Source of Matrix Components Is Essential for De Novo Basement Membrane Formation** *Current Biology: CB* **27**:3526–3534 <https://doi.org/10.1016/j.cub.2017.10.001>
- McNeill H., Yang C. H., Brodsky M., Ungos J., Simon M. A 1997) **Mirror encodes a novel PBX-class homeoprotein that functions in the definition of the dorsal-ventral border in the Drosophila eye** *Genes & Development* **11**:1073–1082 <https://doi.org/10.1101/gad.11.8.1073>
- Moses L., Pachter L 2022) **Museum of spatial transcriptomics** *Nature Methods* **19**:534–546 <https://doi.org/10.1038/s41592-022-01409-2>
- Mu S., Yu S., Turner N. L., McKellar C. E., Dorkenwald S., Collman F., Koolman S., Moore M., Morejohn S., Silverman B., Willie K., Willie R., Bland D., Burke A., Ashwood Z., Luther K., Castro M., Ogedengbe O., Silversmith W., ..., Seung H. S 2021) **3D reconstruction of cell nuclei in a full Drosophila brain** *bioRxiv* <https://doi.org/10.1101/2021.11.04.467197>
- Özel M. N., Simon F., Jafari S., Holguera I., Chen Y.-C., Benhra N., El-Danaf R. N., Kapuralin K., Malin J. A., Konstantinides N., Desplan C 2021) **Neuronal diversity and convergence in a visual system developmental atlas** *Nature* **589**:88–95 <https://doi.org/10.1038/s41586-020-2879-3>
- Pachitariu M., Stringer C 2022) **Cellpose 2.0: How to train your own model** *Nature Methods* **19**:Article 12 <https://doi.org/10.1038/s41592-022-01663-4>
- Partel G., Wählby C 2021) **Spa2vec: Unsupervised representation of localized spatial gene expression signatures** *The FEBS Journal* **288**:1859–1870 <https://doi.org/10.1111/febs.15572>

- Pech U., Janssens J., Schoovaerts N., Kuenen S., Makhzami S., Hulselmans G., Poovathingal S., Bademosi A. T., Swerts J., Vilain S., Aerts S., Verstreken P 2024) **Synaptic deregulation of cholinergic projection neurons causes olfactory dysfunction across 5 fly Parkinsonism models** *eLife* <https://doi.org/10.7554/eLife.98348.1>
- Pinheiro H., Pimentel M. R., Sequeira C., Oliveira L. M., Pezzarossa A., Roman W., Gomes E. R 2021) **mRNA distribution in skeletal muscle is associated with mRNA size** *Journal of Cell Science* **134**:jcs256388 <https://doi.org/10.1242/jcs.256388>
- Pownall M. E., Miao L., Vejnar C. E., M'Saad O., Sherrard A., Frederick M. A., Benitez M. D. J., Boswell C. W., Zaret K. S., Bewersdorf J., Giraldez A. J 2023) **Chromatin expansion microscopy reveals nanoscale organization of transcription and chromatin** *Science* **381**:92–100 <https://doi.org/10.1126/science.ade5308>
- Rodriguez J. M., Rodriguez-Rivas J., Di Domenico T., Vázquez J., Valencia A., Tress M. L. 2018) **APPRIS 2017: Principal isoforms for multiple gene sets** *Nucleic Acids Research* **46**:D213–D217 <https://doi.org/10.1093/nar/gkx997>
- Rodrigues S. G., Stickels R. R., Goeva A., Martin C. A., Murray E., Vanderburg C. R., Welch J., Chen L. M., Chen F., Macosko E. Z 2019) **Slide-seq: A scalable technology for measuring genome-wide expression at high spatial resolution** *Science* **363**:1463–1467 <https://doi.org/10.1126/science.aaw1219>
- Rudolph F., Hüttemeister J., da Silva Lopes K., Jüttner R., Yu L., Bergmann N., Friedrich D., Preibisch S., Wagner E., Lehnart S. E., Gregorio C. C., Gotthardt M. 2019) **Resolving titin's lifecycle and the spatial organization of protein turnover in mouse cardiomyocytes** *Proceedings of the National Academy of Sciences* **116**:25126–25136 <https://doi.org/10.1073/pnas.1904385116>
- Sanes J. R., Johnson Y. R., Kotzbauer P. T., Mudd J., Hanley T., Marttnou J.-C., Merlie J. P 1991) **Selective expression of an acetylcholine receptor-lacZ transgene in synaptic nuclei of adult muscle fibers** *Development* **113**:1181–1191 <https://doi.org/10.1242/dev.113.4.1181>
- Sarov M., Barz C., Jambor H., Hein M. Y., Schmied C., Suchold D., Stender B., Janosch S., Kj V. V., Krishnan R., Krishnamoorthy A., Ferreira I. R., Ejsmont R. K., Finkl K., Hasse S., Kämpfer P., Plewka N., Vinis E., Schloissnig S., ..., Schnorrer F 2016) **A genome-wide resource for the analysis of protein localisation in Drosophila** *eLife* **5**:e12068 <https://doi.org/10.7554/eLife.12068>
- Schaum N., Karkanas J., Neff N. F., May A. P., Quake S. R., Wyss-Coray T., Darmanis S., Batson J., Botvinnik O., Chen M. B., Chen S., Green F., Jones R. C., Maynard A., Penland L., Pisco A. O., Sit R. V., Stanley G. M., Webber J. T., ..., Principal investigators 2018) **Single-cell transcriptomics of 20 mouse organs creates a Tabula Muris** *Nature* **562**:Article 7727 <https://doi.org/10.1038/s41586-018-0590-4>
- Schlegel P., Yin Y., Bates A. S., Dorkenwald S., Eichler K., Brooks P., Han D. S., Gkantia M., Dos Santos M., Munnely E. J., Badalamente G., Serratosa Capdevila L., Sane V. A., Fragniere A. M. C., Kiassat L., Pleijzier M. W., Stürner T., Tamimi I. F. M., Dunne C. R., ..., Jefferis G. S. X. E 2024) **Whole-brain annotation and multi-connectome cell typing of Drosophila** *Nature* **634**:139–152 <https://doi.org/10.1038/s41586-024-07686-5>
- Schönbauer C., Distler J., Jährling N., Radolf M., Dodt H.-U., Frasch M., Schnorrer F 2011) **Spalt mediates an evolutionarily conserved switch to fibrillar muscle fate in insects** *Nature* **479**:Article 7373 <https://doi.org/10.1038/nature10559>

- Schueder F., Mangeol P., Chan E. H., Rees R., Schünemann J., Jungmann R., Görlich D., Schnorrer F 2023) **Nanobodies combined with DNA-PAINT super-resolution reveal a staggered titin nanoarchitecture in flight muscles** *eLife* **12**:e79344 <https://doi.org/10.7554/eLife.79344>
- Schuster C. M 2006) **Glutamatergic synapses of Drosophila neuromuscular junctions: A high-resolution model for the analysis of experience-dependent potentiation** *Cell and Tissue Research* **326**:287–299 <https://doi.org/10.1007/s00441-006-0290-5>
- Sênos Demarco R., Stack B. J., Tang A. M., Voog J., Sandall S. L., Southall T. D., Brand A. H., Jones D. L. 2022) **Escargot controls somatic stem cell maintenance through the attenuation of the insulin receptor pathway in Drosophila** *Cell Reports* **39**:110679 <https://doi.org/10.1016/j.celrep.2022.110679>
- Sokolowski M. B 2001) **Drosophila: Genetics meets behaviour** *Nature Reviews Genetics* **2**:Article 11 <https://doi.org/10.1038/35098592>
- Solorzano L., Partel G., Wählby C 2020) **TissUMaps: Interactive visualization of large-scale spatial gene expression and tissue morphology data** *Bioinformatics* **36**:4363–4365 <https://doi.org/10.1093/bioinformatics/btaa541>
- Spletter M. L., Barz C., Yeroslaviz A., Zhang X., Lemke S. B., Bonnard A., Brunner E., Cardone G., Basler K., Habermann B. H., Schnorrer F 2018) **A transcriptomics resource reveals a transcriptional transition during ordered sarcomere morphogenesis in flight muscle** *eLife* **7**:e34058 <https://doi.org/10.7554/eLife.34058>
- Stahl A. L., Charlton-Perkins M., Buschbeck E. K., Cook T. A 2017) **The cuticular nature of corneal lenses in Drosophila melanogaster** *Development Genes and Evolution* **227**:271–278 <https://doi.org/10.1007/s00427-017-0582-7>
- Steib E., Tetley R., Laine R. F., Norris D. P., Mao Y., Vermot J 2022) **TissUExM enables quantitative ultrastructural analysis in whole vertebrate embryos by expansion microscopy** *Cell Reports Methods* **2**:100311 <https://doi.org/10.1016/j.crmeth.2022.100311>
- Stirling D. R., Swain-Bowden M. J., Lucas A. M., Carpenter A. E., Cimini B. A., Goodman A 2021) **CellProfiler 4: Improvements in speed, utility and usability** *BMC Bioinformatics* **22**:433 <https://doi.org/10.1186/s12859-021-04344-9>
- The Tabula Sapiens consortium. 2022) **The Tabula Sapiens: A multiple-organ, single-cell transcriptomic atlas of humans** *Science* **376**:eabl4896 <https://doi.org/10.1126/science.abl4896>
- Tomancak P., Beaton A., Weiszmam R., Kwan E., Shu S., Lewis S. E., Richards S., Ashburner M., Hartenstein V., Celniker S. E., Rubin G. M 2002) **Systematic determination of patterns of gene expression during Drosophila embryogenesis** *Genome Biology* **3**:research0088.1 <https://doi.org/10.1186/gb-2002-3-12-research0088>
- Wang M., Hu Q., Lv T., Wang Y., Lan Q., Xiang R., Tu Z., Wei Y., Han K., Shi C., Guo J., Liu C., Yang T., Du W., An Y., Cheng M., Xu J., Lu H., Li W., ..., Liu L 2022) **High-resolution 3D spatiotemporal transcriptomic maps of developing Drosophila embryos and larvae** *Developmental Cell* **57**:1271–1283 <https://doi.org/10.1016/j.devcel.2022.04.006>
- Wolf F. A., Angerer P., Theis F. J 2018) **SCANPY: Large-scale single-cell gene expression data analysis** *Genome Biology* **19**:15 <https://doi.org/10.1186/s13059-017-1382-0>

Yao Z., Velthoven C. T. J. van, Kunst M., Zhang M., McMillen D., Lee C., Jung W., Goldy J., Abdelhak A., Baker P., Barkan E., Bertagnolli D., Campos J., Carey D., Casper T., Chakka A. B., Chakrabarty R., Chavan S., Chen M., ..., Zeng H. 2023) **A high-resolution transcriptomic and spatial atlas of cell types in the whole mouse brain** *bioRxiv* <https://doi.org/10.1101/2023.03.06.531121>

Yarnitzky T., Volk T 1995) **Laminin Is Required for Heart, Somatic Muscles, and Gut Development in the Drosophila Embryo** *Developmental Biology* **169**:609–618 <https://doi.org/10.1006/dbio.1995.1173>

Yoo S., Nair S., Kim H., Kim Y., Lee C., Lee G., Park J. H 2020) **Knock-in mutations of scarecrow, a Drosophila homolog of mammalian Nkx2.1, reveal a novel function required for development of the optic lobe in Drosophila melanogaster** *Developmental Biology* **461**:145–159 <https://doi.org/10.1016/j.ydbio.2020.02.008>

Zhang M., Eichhorn S. W., Zingg B., Yao Z., Cotter K., Zeng H., Dong H., Zhuang X 2021) **Spatially resolved cell atlas of the mouse primary motor cortex by MERFISH** *Nature* **598**:Article 7879 <https://doi.org/10.1038/s41586-021-03705-x>

Author information

Jasper Janssens

VIB-KU Leuven Center for Brain and Disease Research, KU Leuven, Leuven, Belgium, Laboratory of Computational Biology, Department of Human Genetics, KU Leuven, Leuven, Belgium
ORCID iD: [0000-0001-9264-0782](https://orcid.org/0000-0001-9264-0782)

Pierre Mangeol[#]

Aix Marseille University, CNRS, IBDM, Turing Centre for Living Systems, Marseille, France
ORCID iD: [0000-0002-8305-7322](https://orcid.org/0000-0002-8305-7322)

[#]Co-2nd authors

Nikolai Hecker[#]

VIB-KU Leuven Center for Brain and Disease Research, KU Leuven, Leuven, Belgium, VIB Center for AI & Computational Biology, KU Leuven, Leuven, Belgium, Laboratory of Computational Biology, Department of Human Genetics, KU Leuven, Leuven, Belgium
ORCID iD: [0000-0003-1693-4257](https://orcid.org/0000-0003-1693-4257)

[#]Co-2nd authors

Gabriele Partel[#]

VIB-KU Leuven Center for Brain and Disease Research, KU Leuven, Leuven, Belgium, VIB Center for AI & Computational Biology, KU Leuven, Leuven, Belgium, Laboratory of Computational Biology, Department of Human Genetics, KU Leuven, Leuven, Belgium

[#]Co-2nd authors

Katina Spanier

VIB-KU Leuven Center for Brain and Disease Research, KU Leuven, Leuven, Belgium, VIB Center for AI & Computational Biology, KU Leuven, Leuven, Belgium, Laboratory of Computational Biology, Department of Human Genetics, KU Leuven, Leuven, Belgium

Joy Ismail

VIB-KU Leuven Center for Brain and Disease Research, KU Leuven, Leuven, Belgium, Laboratory of Computational Biology, Department of Human Genetics, KU Leuven, Leuven, Belgium

Gert Hulselmans

VIB-KU Leuven Center for Brain and Disease Research, KU Leuven, Leuven, Belgium, VIB Center for AI & Computational Biology, KU Leuven, Leuven, Belgium, Laboratory of Computational Biology, Department of Human Genetics, KU Leuven, Leuven, Belgium

Stein Aerts*

VIB-KU Leuven Center for Brain and Disease Research, KU Leuven, Leuven, Belgium, VIB Center for AI & Computational Biology, KU Leuven, Leuven, Belgium, Laboratory of Computational Biology, Department of Human Genetics, KU Leuven, Leuven, Belgium
ORCID iD: [0000-0002-8006-0315](https://orcid.org/0000-0002-8006-0315)

For correspondence: stein.aerts@kuleuven.be

* Co-last authors

Frank Schnorrer*

Aix Marseille University, CNRS, IBDM, Turing Centre for Living Systems, Marseille, France
ORCID iD: [0000-0002-9518-7263](https://orcid.org/0000-0002-9518-7263)

For correspondence: frank.schnorrer@univ-amu.fr

* Co-last authors

Editors

Reviewing Editor

Hugo Bellen

Baylor College of Medicine, Houston, United States of America

Senior Editor

Claude Desplan

New York University, New York, United States of America

Reviewer #1 (Public review):

Summary:

In this manuscript, Janssens et al. addressed the challenge of mapping the location of transcriptionally unique cell types identified by single nuclei sequencing (snRNA-seq) data available through the Fly Cell Atlas. They identified 100 transcripts for head samples and 50 transcripts for fly body samples allowing identification of every unique cell type discovered through the Fly Cell Atlas. To map all of these cell types, the authors divided the fly body into head and body samples and used the Molecular Cartography (Resolve Biosciences) method to

visualize these transcripts. This approach allowed them to build spatial tissue atlases of the fly head and body, to identify the location of previously unknown cell types and the subcellular localization of different transcripts. By combining snRNA-seq data from the Fly Cell Atlas with their spatially resolved transcriptomics (SRT) data, they demonstrated an automated cell type annotation strategy to identify uncharacterized clusters and infer their location in the fly body. This manuscript constitutes a proof-of-principle study to map the location of the cells identified by ever-growing single-cell transcriptomics datasets generated by others.

Strengths:

The authors used the Molecular Cartography (Resolve Biosciences) method to visualize 100 transcripts for head samples and 50 transcripts for fly body samples in high resolution. This method achieves high resolution by multiplexing a large number of transcript visualization steps and allows the authors to map the location of unique cell types identified by the Fly Cell Atlas.

Weaknesses:

Combining single-nuclei sequencing (snRNA-seq) data with spatially resolved transcriptomics (SRT) data is challenging, and the methods used by the authors in this study cannot reliably distinguish between cells, especially in brain regions where the processes of different neurons are clustered, such as neuropils. This means that a grid that the authors mark as a unique cell may actually be composed of processes from multiple cells.

Comments on revisions:

I believe the authors have improved the manuscript by addressing all the concerns and incorporating the suggestions raised by the reviewers. I have no further concerns or suggestions.

<https://doi.org/10.7554/eLife.92618.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Public Reviews:

Reviewer #1 (Public Review):

Summary:

In this manuscript, Janssens et al. addressed the challenge of mapping the location of transcriptionally unique cell types identified by single nuclei sequencing (snRNA-seq) data available through the Fly Cell Atlas. They identified 100 transcripts for head samples and 50 transcripts for fly body samples allowing the identification of every unique cell type discovered through the Fly Cell Atlas. To map all of these cell types, the authors divided the fly body into head and body samples and used the Molecular Cartography (Resolve Biosciences) method to visualize these transcripts. This approach allowed them to build spatial tissue atlases of the fly head and body, to identify the location of previously unknown cell types and the subcellular localization of different transcripts. By combining snRNA-seq data from the Fly Cell Atlas with their spatially resolved transcriptomics (SRT) data, they demonstrated an automated cell type annotation strategy to identify unknown clusters and infer their location in the fly body. This manuscript constitutes a proof-of-

principle study to map the location of the cells identified by ever-growing single-cell transcriptomic datasets generated by others.

Strengths:

The authors used the Molecular Cartography (Resolve Biosciences) method to visualize 100 transcripts for head samples and 50 transcripts for fly body samples in high resolution. This method achieves high resolution by multiplexing a large number of transcript visualization steps and allows the authors to map the location of unique cell types identified by the Fly Cell Atlas.

We thank this reviewer for appreciating the quality of our spatial data. We do not know what caused the technical problem (grayscale version of PDF) for this reviewer (the PDF figures are in color on the eLife website and on bioRxiv). We are surprised that the eLife discussion session did not resolve this issue.

Weaknesses:

Combining single-nuclei sequencing (snRNA-seq) data with spatially resolved transcriptomics (SRT) data is challenging, and the methods used by the authors in this study cannot reliably distinguish between cells, especially in brain regions where the processes of different neurons are clustered, such as in neuropils. This means that a grid that the authors mark as a unique cell may actually be composed of processes from multiple cells.

The small size of an individual fly is one of the most challenging aspects of performing spatial transcriptomics. While the resolution of Molecular Cartography is rather high (< 200 nm), in the brain challenges remain as noted by the reviewer. *Drosophila* neuronal nuclei are notoriously small and cannot be easily resolved with the current imaging techniques. We agree that for a full atlas either expansion microscopy, 3D techniques or other super-resolution techniques will be required.

Reviewer #2 (Public Review):

Summary:

The landmark publication of the "Fly Atlas" in 2022 provided a single cell/nuclear transcriptomic dataset from 15 individually dissected tissues, the entire head, and the body of male and female flies. These data led to the annotation of more than 250 cell types. While certainly a powerful and datarich approach, a significant step forward relies on mapping these data back to the organism in time and space. The goal of this manuscript is to map 150 transcripts defined by the Fly Atlas by FISH and in doing so, provide, for the first time, a spatial transcriptomic dataset of the adult fly. Using this approach (Molecular Cartography with Resolve Biosciences), the authors, furthermore, distinguish different RNA localizations within a cell type. In addition, they seek to use this approach to define previously unannotated clusters found in the Fly Atlas. As a resource for the community at large interested in the computational aspects of their pipeline, the authors compare the strengths and weaknesses of their approach to others currently being performed in the field.

Strengths:

(1) The authors use Resolve Biosciences and a novel bioinformatics approach to generate a FISHbased spatial transcriptomics map. To achieve this map, they selected 150 genes (50 body; 100 head) that were highly expressed in the single nuclear RNA sequencing dataset and were used in the 2022 paper to annotate specific cell types; moreover, the

authors chose several highly expressed genes characteristic of unannotated cell types. Together, the approach and generated data are important next steps in translating the transcriptomic data to spatial data in the organism.

We thank the reviewer for this comment, as it reminded us that we need to be clearer in the text, about how we chose the genes to investigate. The statement that we selected “150 genes (50 body; 100 head) that were highly expressed in the single nuclear RNA sequencing dataset” is not correct. We have chosen genes with widely differing expression levels (log-scale range of 3.95 in body, 5.76 in head, we show this now in the new Figure 1 – figure supplement 1B, D). Many of the chosen genes are also transcription factors. In fact, the here introduced method is more sensitive than the single cell atlas: the tinman positive cells were readily located (even non-heart cells were found to express tinman), whereas in the single cell FCA data tinman expression is often not detected in the cardiomyocytes (tinman is detected in 273 cells in the entire FCA (mean expression of 1.44 UMI in positive cells), and in 71 cells out of 273 cardiac cells (26%)).

(2) Working with Resolve, the authors developed a relatively high throughput approach to analyze the location of transcripts in Drosophila adults. This approach confirmed the identification of particular cell types suggested by the FlyAtlas as well as revealed interesting subcellular locations of the transcripts within the cell/tissue type. In addition, the authors used co-expression of different RNAs to unbiasedly identify "new cell types". This pipeline and data provide a roadmap for additional analyses of other time points, female flies, specific mutants, etc.

(3) The authors show that their approach reveals interesting patterns of mRNA distribution (e.g alpha- and beta-Trypsin in apical and basal regions of gut enterocytes or striped patterns of different sarcomeric proteins in body muscle). These observations are novel and reveal unexpected patterns. Likewise, the authors use their more extensive head database to identify the location of cells in the brain. They report the resolution of 23 clusters suggested by the single-cell sequencing data, given their unsupervised clustering approach. This identification supports the use of spatial cell transcriptomics to characterize cell types (or cell states).

(4) Lastly, the authors compare three different approaches --- their own described in this manuscript, Tangram, and SpaGE - which allow integration of single cell/nuclear RNA-seq data with spatial localization FISH. This was a very helpful section as the authors compared the advantages and disadvantages (including practical issues, like computational time).

Weaknesses:

(1) Experimental setup. It is not clear how many and, for some of the data, the sex of the flies that were analyzed. It appears that for the body data, only one male was analyzed. For the heads, methods say male and female heads, but nothing is annotated in the figures. As such, it remains unclear how robust these data are, given such a limited sample from one sex. As such, the claims of a spatial atlas of the entire fly body and its head ("a rosetta stone") are overstated. Also, the authors should clearly state in the main text and figure legends the sex, the age, how many flies, and how many replicates contributed to the data presented (not just the methods). What also adds to the confusion is the use of "n" in para 2 of the results. "... we performed coronal sections at different depths in the head (n=13)..." 13 sections in total from 1 head or sections from 13 heads? Based on the body and what is shown in the figure, one assumes 13 sections from one head. Please clarify.

While we agree that sex differences present indeed an interesting opportunity to study with spatial transcriptomics, our goal was not to define male/female differences but rather to establish the technology to go into this detail if wanted in the future. In the revised version, we have provided an additional supplementary table with a more detailed description of the head sections (Table S3). We have added the number of animals (12 for the head sections, mixed sex; and 1 male for the body sections) to the main text. We would like to point out that we verified the specificity of our MC method on all the 5 body sections (Figure 2A, TpnC4 & Act88F and text) and not only on one. Furthermore, we also would like to state that the idea of “a Rosetta stone” was mentioned as a future prospect that clearly goes beyond our presented work. We have rewritten the discussion to make this clearer and to any avoid overstatements.

(2) Probes selected: Information from the methods section should be put into the main text so that it is clear what and why the gene lists were selected. The current main text is confusing. If the authors want others to use their approach, then some testing or, at the very least, some discussion of lower expressed genes should be added. How useful will this approach be if only highly expressed genes can be resolved? In addition, while it is understood that the company has a propriety design algorithm for the probes, the authors should comment on whether the probes for individual genes detect all isoforms or subsets (exons and introns?), given the high level of splicing in tissues such as muscle.

As stated above, while there is a slight bias to higher expressed genes (as expected for marker genes), we have also used low expressed genes like salm, CG32121, tinman (body) or sens (head). This is now shown in new Figure 1 – figure Supplement 1B, D. This shows that our method is more sensitive than single-cell data, as all cardiomyocytes can be identified by tinman expression and not only some are positive, as is the case in the FCA data. In fact, the method cannot resolve too highly expressed genes due to optical crowding of the signal leading to a worse quantification. For this reason, ninaE was removed from the analysis (as mentioned in Spatial transcriptomics allows the localization of cell types in the head and brain and in Methods).

As mentioned in the Methods, the probes are designed on gene level targeting all isoforms, but favoring principal isoforms (weighted by APPRIS level). The high level of splicing is indeed interesting and we expect that in the future spatial transcriptomics can help to generate more insight into this by designing isoform-specific probes.

(3) Imaging: it isn't clear from the text whether the repeated rounds of imaging impacted data collection. In many of what appear to be "stitched" images, there are gradients of signal (eg, figure 2F); please comment. Also, since this a new technique, could a before and after comparison of the original images and the segmented images be shown in the supplemental data so that the reader can better appreciate how the authors assessed/chose/thresholded their data? More discussion of the accuracy of spot detection would be helpful.

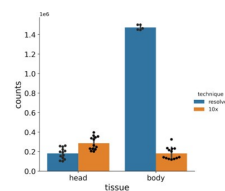
High-resolution imaging (pixel size = 138 nm) of a large field of view (>1mm) for spatial transcriptomics uses a stitching method to combine several individual images to reconstruct a large field of view. This does not generate signal gradients, apart from lower signal at the extreme edges of each of the individual images, as seen in our images, too. The spot detection algorithm was written and used by Resolve Biosciences and benchmarked for human (Hela) and mouse (NIH-3T3) cell lines in Groiss et al. 2021 (Highly resolved spatial transcriptomics for detection of rare events in cells, bioRxiv). The specificity of the decoded probes was found to lie between 99.45 and 99.9% here, matching the results we found for specific detection of TpnC4 and Act88F (99.4 and 99.8%).

(4) The authors comment on how many RNAs they detected (first paragraph of results). How do these numbers compare to the total mRNA present as detected by single-cell or single-nuclear sequencing?

We can compare the numbers, but the different methodologies make the interpretation of such a comparison difficult. FCA used single nucleus sequencing, so only nuclear pre-mRNAs are detected. The total amount of counts per single cell sample strongly depends on how many cells were sequenced in an experiment. MC detects all mRNAs present in the section. Here, the size of the sample and hence the size or the number of cells analyzed determines how many mRNAs are detected. In Author response image 1, we have compared our MC results versus FCA data, comparing the genes investigated here in MC per section vs per sequencing experiment. Numbers for MC are slightly lower for the brain (not all cell types are on all sections) and much higher for the larger body samples. However, we feel a direct comparison is questionable, so we prefer to not include this figure in our manuscript.

Author response image 1:

Barplots showing total number of mRNA molecules detected in Molecular Cartography (MC, Resolve, spatial spots) and in snRNA-seq data from the Fly Cell Atlas (10x Genomics, UMIs). Individual black dots show individual experiments, counts are only shown for the chosen gene panel for each sample. Bar shows the mean, with error bars representing the standard error.



(5) Using this higher throughput method of spatial transcriptomics, the authors discern different cell types and different localization patterns within a tissue/cell type.

a. The authors should comment on the resolution provided by this approach, in terms of the detection of populations of mRNAs detected by low throughput methods, for example, in glia, motor neuron axons, and trachea that populate muscle tissue. Are these found in the images? Please show.

We did not add any markers for trachea in our gene panel, but we do detect sparse spots of repo (glia) and elav/VGluT in the muscle tissues (Gad1/VACHT are hardly detected in the muscle tissue). This is consistent with the glutamatergic nature of motor neurons in Drosophila as described previously (Schuster CM (2006), Glutamatergic synapses of Drosophila neuromuscular junctions: a high-resolution model for the analysis of experience-dependent potentiation. Cell Tissue Res 326:

287–299.). We have present these new data in new Figure 2 – figure supplement 1.

b. The authors show interesting localization patterns in muscle tissue for different sarcomere protein-coding mRNAs, including enrichment of sls in muscle nuclei located near the muscle-tendon attachment sites. As this high throughput approach is newly being applied to the adult fly, it would increase confidence in these data, if the authors would confirm these data using a low throughput FISH technique. For example, do the authors detect such alternating "stripes" (Act 88F, TpnC4, and Mhc) or enriched

localization (sls) using FISH that doesn't rely on the repeated colorization, imaging, decolorization of the probes?

We thank the reviewer for the interest in the localization patterns in muscle tissue. We show that Act88F, TpnC4 are not detected outside of flight muscle cells (99.4% and 99.8% of the single molecular signal in flight muscles only), giving us confidence in the specificity of the MC method. Following the suggestion of the reviewer, we have adapted an HCR-FISH method to *Drosophila* adult body sections for the revised version of the manuscript. Using this method, we were able to confirm the higher expression/localization of sls transcripts to and around the adult flight muscle nuclei, with an enrichment in nuclei close to the muscle-tendon attachment sites (new Figure 4D-F and new Figure 4 – figure supplement 1). We have also been able to confirm some complementarity in the localization patterns of Act88F and TpnC4 in longitudinal stripes in adult flight muscles, however for Mhc we could not confirm this pattern with HCR-FISH (new Figure 5C-F and new Figure 5 – figure supplement 1). While we could confirm most of the pattern seen, we do not know the exact reason for the slight discrepancies. Thus, we now recommend that insights found with SRT should be confirmed with more classical FISH methods.

(6) The authors developed an unbiased method to identify "new cell types" which relies on coexpression of different transcripts. Are these new cell types or a cell state? While expression is a helpful first step, without any functional data, the significance of what the authors found is diminished. The authors need to soften their statements.

The term “new cell types” only appeared in the old title. We agree that with the current spatial map we cannot be sure to have found “new cell types”, instead we show where unannotated/uncharacterized clusters from the scRNA-seq atlas are located, based on their gene expression. Therefore, we have updated the title in the revised version (Spatial transcriptomics in the adult *Drosophila* brain and body) and thank the reviewer for this valuable suggestion.

Appraisal:

The authors' goal is to map single cell/nuclear RNAseq data described in the 2022 Fly Atlas paper spatially within an organism to achieve a spatial transcriptomic map of the adult fly; no doubt, this is a critical next step in our use of 'omics approaches. While this manuscript does the hard work of trying to take this next step, including developing and testing a new pipeline for high throughput FISH and its analysis, it falls short, in its present form, in achieving this goal. The authors discuss creating a robust spatial map, based on one male fly. Moreover, they do not reveal principles of mRNA localization, as stated in the abstract; they show us patterns, but nothing about the logic or function of these patterns. This same criticism can be said of the identification of "new cell types, just based on RNA colocalization. In both cases (mRNA subcellular localization or cell type identification), further data in the form of validation with traditional low throughput FISH and genetic manipulations to assess the relation to cell function are required for the authors to make such claims.

We have indeed used one male fly for the adult male body data. This is mainly due to the cost of the sample processing. We used 12 individuals for the head samples (from 1 individual we acquired 2 sections, a total of 13 sections). We show that the body samples show a high correlation with each other, while the head samples cover multiple depths of the head. Still, even in the head, we find that sections at similar depths show a high similarity to each other in terms of gene-gene coexpression and expression patterns. Although obtaining sections from more animals would be valuable, we do not believe it to be necessary for our current goals. Additional replicates beyond the ones we already provide would require significant

amounts of extra time and budget, while they would very likely produce similar results as we already show. Following the reviewer's suggestion, we have tested several genes with HCR-FISH and could readily confirm the localization pattern of sls mRNA close to the terminal nuclei of the flight muscles. This pattern is likely due to a higher expression of sls in these nuclei, as a large amount of sls mRNA signal is detected within the nuclei (Figure 4). A detailed dissection of the mechanism that establishes this pattern is beyond the scope of this manuscript, which is the first one on applying spatial transcriptomics to adult *Drosophila*.

The usage of the term "new cell types" was indeed ambiguous and we removed this from the revised version. We now clarified that we map the spatial location of unannotated clusters in the brain. This may or may not include uncharacterized cell types. We now further specify that we have only inferred the location of the nuclei; thus, neuronal function or the location of their axonal processes are still unknown. As such, our data provides a starting point to identify uncharacterized cell types, since their marker genes and nuclear location are now determined. The next step to identify "new cell types" would indeed be to acquire genetic access to these cell types and characterize them in more detail. This is beyond the scope of this manuscript, and therefore we have toned down the title in the revised version and thank the reviewer for this valuable suggestion.

Discussion of likely impact:

If revised, these data, and importantly the approach, would impact those working on Drosophila adults as well as those working in other model systems where single cell/nuclear sequencing is being translated to the spatial localization within the organism. The subcellular localization data - for example, the size of transcripts and how that relates to localization or the patterns of sarcomeric protein localization in muscle - are intriguing, and would likely impact our thinking on RNA localization, transport, etc if confirmed. Lastly, the authors compare their computational approaches to those available in the field; this is valuable as this is a rapidly evolving field and such considerations are critical for those wishing to use this type of approach.

We thank this reviewer for appreciating the impact of our findings and approach to the *Drosophila* field and beyond. We here provide the groundwork for a full *Drosophila* adult spatial atlas, similar to how early scRNA-seq datasets provided a framework for the Fly Cell Atlas. In the manuscript we provide both experimental information on how to successfully perform spatial transcriptomics (treating slides for optimal attachment) and the data serves as a benchmark for future experiments to improve upon (similar to how early Drop-seq datasets were compared to later 10x datasets in single-cell transcriptomics). In addition, it also provides proof of principle methods on how to integrate the FCA data with these spatial data and it identifies localized mRNA species in large adult muscle cells, showing the complementarity of spatial techniques with single-cell RNA-seq. For a small number of genes, we have confirmed the mRNA patterns using HCR-FISH in the revised version of this manuscript. To conclude, this is the first spatial adult *Drosophila* transcriptomics paper, locating 150 mRNA species with easy data access in our user portal (<https://spatialfly.aertslab.org/>).

Recommendations for the authors:

Reviewer #1 (Recommendations For The Authors):

(1) All figures in the manuscript were in grayscale, which made it difficult to interpret the results because the data could only be interpreted by distinguishing different colors to visualize different transcripts. This is likely a technical problem. The manuscript must contain colored images.

We apologize to the reviewer for this technical issue. The manuscript was uploaded in color to bioRxiv and to eLife. We therefore do not understand to reason for this problem. We are surprised that this issue was not resolved in the reviewers' discussion since color is obviously essential to appreciate the beauty of this manuscript.

(2) In Figure 2a, the authors comment on the subcellular localization of trypsin isoforms, but the figure does not indicate the cell borders or the apical and basal regions of the cell. These must be indicated in the figure to help readers understand the results.

We thank the reviewer for pointing this out; we have now indicated the outlines of the single-cell layer epithelium on the figure. While we have no marker for cell borders, we have a nuclear marker showing that it is a single cell layer. We hope this allows the reader to appreciate the subcellular localization of the trypsin isoforms.

(3) All figures (including the data on the authors' website) contain background staining, which I assume is labeling nuclei. This is not indicated in the manuscript, and should be clarified.

We again thank the reviewer for pointing this out; the background staining indeed labels nuclei (using DAPI). We have indicated this better in the revised version.

(4) In Figure 5c, the authors claim that neuronal and muscular genes are grouped into the same cluster, but they do not indicate which transcripts are neuronal and which ones are muscular. This must be indicated in the figure.

We thank the reviewer for this comment. Indeed, there was only one gene, *acj6*, present in the muscle cluster. So, we decided to delete this statement in the revised version.

(5) The authors utilized and compared three different approaches to integrate single nuclei sequencing data from the Fly Cell Atlas to their spatially resolved transcriptomics (SRT) data. I was wondering if it is possible to generate a virtual expression explorer using this integrated data, similar to the dataset published in the 2017 Science article by Karaïskos et al., where they combined publicly available in situ hybridization data of fly embryos and their single-cell sequencing data. This virtual expression explorer would be useful to visualize the expression pattern of transcripts that the authors of this paper did not use for their SRT.

We thank the reviewer for this interesting comment. Using Tangram, we indeed infer gene expression for all genes from the Fly Cell Atlas. To make this visible we have created a Scope session (https://scope.aertslab.org/#/Spatial_Fly*/welcome), with which users can browse inferred gene expression levels (note that this is on a segmented cell level). We do notice that the inferred gene expression levels contain many false positives and should therefore be used with caution. The spatial data themselves can be browsed through the spatial portal at <https://spatialfly.aertslab.org/>.

Reviewer #2 (Recommendations For The Authors):

Suggestions for improved or additional experiments, data, or analyses:

The authors have used a new high throughput approach to examine the location of 150 RNAs in adult Drosophila heads or one body. It is unclear whether the fixation/repeated imaging etc is accurately reflecting the patterns of expression in vivo. The authors should confirm these data using low throughput established techniques for the RNA patterns in muscle for example.

The authors should clarify their experimental approaches and include additional samples if they indeed want to establish the rosetta stone of fly adults. These data are from only a male fly (and as such is not a complete analysis of the adult fly). To be a map of the adult fly, data from both sexes need to be included.

Unless functional data that complement the descriptive data shown here are included, the authors have to soften their conclusions. For example, while spatial transcriptomics has mapped RNA expression to a location, without some functional data, it is difficult to conclude that these are indeed "new cell types". Same with the RNA localization principles.

Recommendations for improving the writing and presentation:

(1) The manuscript should be heavily revised: in many places, important details are left out or should be moved from the methods to the main text. In addition, the authors often overstate their findings throughout the manuscript. As an example, it appears that the data presented is only from 1 fly, so this doesn't increase the reader's confidence in the data or the applicability of the approach. Also, it isn't clear how many flies were analyzed for the heads (one male fly too?) nor what variability is present from fly to fly. For the approach and data to be used by others, this is important to know.

We moved some text from the methods section to the main text to be clearer. We now also state how many animals were used for the MC method. While the data for the body has been generated from 1 male only, the data for the head was generated from 12 flies; for both cases, similar slices show very similar gene expression patterns. Furthermore, in the body we used widely known and published marker genes that all showed expected expression patterns, indicating robustness. We agree that this is not a full spatial atlas of the fly, this was also not our goal and we have removed such general statements from the revised version: we aimed to generate a spatial transcriptomics dataset, covering the entire fly (head and body) as a proof-of-principle, tackling data generation and analysis, and highlighting challenges in both.

(2) The grammar and word choice throughout are challenging often making the text difficult to follow. This reads like an early draft of the paper.

We apologize to the reviewer for any difficulties. We have revised the text and hope it is now easier to read, while still being accurate on the technical details of the various methods used in our manuscript.

Minor corrections to the text and figures.

See the weaknesses mentioned above. Also:

Figure S1 is unreadable.

There is no simple way to describe the expression values of 100 genes in 100 cell types on a single page. The resolution of the PDF is high enough that after zooming in, all the information can be read easily.

Figure S2, in a, please include the axes so that the reader can better understand the sections shown.

In b, it is unclear what the pink boxes mean. In c, the labels are barely legible.

In Figure 1 – figure supplement 2 (head sections), we have ordered the head sections from anterior to posterior. The boxes in (B) represent boxplots. We have updated this plot for

clarity to better display the number of mRNA molecules detected for each gene. We have increased the font size in (C).

| *Figure S3, in a, please include axes. In b, the meaning of the pink box*

In Figure 1 – figure supplement 3 (the body sections) we have added the anterior to posterior and dorso-ventral axis, and ordered the sections that stem from the same animal. The boxes in (B) represent boxplots. We have updated this plot for clarity to better display the number of mRNA molecules detected for each gene. We have added an explanation to the figure legend.

| *Figure S4, the text in the axes of the heatmap should have a darker typeface*

We have changed it to black, thanks.

| *Figure S5c, are the colors in the dendrogram supposed to match the spatial location on the right?*

| *The purple of the muscles is barely visible.*

Yes, they do match. Colors were modified in the revised version for better visibility.

<https://doi.org/10.7554/eLife.92618.2.sa0>